

Wider mathematics education corpus (N = 568)

About this list

Supplementary list for Pampaka, M., Zhang, C., Black, L., Choudry, S., & Harris, D. (upcoming). From gender gaps to gendered participation: Four decades of research in mathematics education (1985-2025).

Records were retrieved from Scopus in January 2026, using the chapter's gender-related search terms with the journal name restricted to titles containing "mathematic*" OR "Statistics" AND "education" OR "teach*" OR "learn*" OR "school". The search returned 1,342 entries. After title and abstract screening against the chapter's inclusion criteria, 568 publications were retained. This corpus was used only for the topic modelling comparison reported in the chapter, and not for the thematic analysis or the citation analysis.

Bibliographic details were completed from Crossref and OpenAlex by DOI, and accepted only where the retrieved title matched the Scopus record. Records without a usable DOI were completed from Scopus directly. Titles appear as published, so capitalisation is not uniform.

References

- Abdalla, S., Alhaj, A., & Ramadan, E. (2025). Motivational factors for visually impaired college students in learning data analytics. *Eurasia Journal of Mathematics, Science and Technology Education*, 21(2), em2578. <https://doi.org/10.29333/ejmste/15923>
- Abdullah, A. H., Mokhtar, M., Abd Halim, N. D., Ali, D. F., Mohd Tahir, L., & Abdul Kohar, U. H. (2016). Mathematics Teachers' Level of Knowledge and Practice on the Implementation of Higher-Order Thinking Skills (HOTS). *EURASIA Journal of Mathematics, Science and Technology Education*, 13(1). <https://doi.org/10.12973/eurasia.2017.00601a>
- Abuhasanein, Y., Rejeb, A., & Jemai, K. (2025). Challenges to learning mathematical concepts among sixth-grade students in primary education: A teachers' perspective. *International Electronic Journal of Mathematics Education*, 20(2), em0818. <https://doi.org/10.29333/iejme/15918>
- Adiredja, A. P. (2019). Anti-Deficit Narratives: Engaging the Politics of Research on Mathematical Sense Making. *Journal for Research in Mathematics Education*, 50(4), 401–435. <https://doi.org/10.5951/jresmetheduc.50.4.0401>
- Agirre-Basurko, E., Albizu-Mallea, U., & Perales-Fernandez-de-Gamboa, A. (2024). Athletes as guest speakers in mathematics education: A descriptive study of a pala player in dialogue with pre-service teachers. *Journal on Mathematics Education*, 15(2), 683–698. <https://doi.org/10.22342/jme.v15i2.pp683-698>

- Aguilar, M. S., Rosas, A., Zavaleta, J. G. M., & Romo-Vázquez, A. (2016). EXPLORING HIGH-ACHIEVING STUDENTS' IMAGES OF MATHEMATICIANS. *International Journal of Science and Mathematics Education*, 14(3), 527–548. <https://doi.org/10.1007/s10763-014-9586-1>
- Agwu, N., Edema, M., Ogunsuyi, O., Williams, O., Williams, S., & Bello, A. (2015). Culture and women's stories: A framework for capacity building in science, technology, engineering and mathematics (stem) related fields. *Mathematics Teaching-Research Journal*, 7(2).
- Akgunduz, D. (2016). A Research about the Placement of the Top Thousand Students Placed in STEM Fields in Turkey between the years 2000 and 2014. *EURASIA Journal of Mathematics, Science and Technology Education*, 12(5). <https://doi.org/10.12973/eurasia.2016.1518a>
- Akhmetova, A. I., Mukazhanova, R. A., Aitpayeva, A. K., Urazova, M. B., Jamoldinova, O. R., & Khodjaev, B. K. (2025). An examination of pre-service teachers' environmental attitudes, behaviors, and knowledge. *Eurasia Journal of Mathematics, Science and Technology Education*, 21(6), em2643. <https://doi.org/10.29333/ejmste/16415>
- Akimov, A., Malin, M., Sargsyan, Y., Suyunov, G., & Turdaliev, S. (2023). Student Success in a University First-Year Statistics Course: Do Students' Characteristics Affect Their Academic Performance? *Journal of Statistics and Data Science Education*, 32(1), 24–35. <https://doi.org/10.1080/26939169.2023.2184435>
- Akinsola, M. K., & Awofala, A. O. A. (2009). Effect of personalization of instruction on students' achievement and self-efficacy in mathematics word problems. *International Journal of Mathematical Education in Science and Technology*, 40(3), 389–404. <https://doi.org/10.1080/00207390802643169>
- Akın, A., Güzeller, C. O., & Sezer Evcan, S. (2016). The Development of a Mathematics Self-Report Inventory for Turkish Elementary Students. *EURASIA Journal of Mathematics, Science and Technology Education*, 12(9). <https://doi.org/10.12973/eurasia.2016.1264a>
- Al Darayseh, A., & Mersin, N. (2025). Integrating generative AI into STEM education: Insights from science and mathematics teachers. *International Electronic Journal of Mathematics Education*, 20(3), em0832. <https://doi.org/10.29333/iejme/16232>
- AL-Azzam, M. M., Hamadneh, B. M., Alqarni, T. M., & Almalki, A. D. (2023). The Degree of Using Blended Learning among Teachers of Students with Learning Disabilities. *International Journal of Education in Mathematics, Science and Technology*, 11(6), 1443–1457. <https://doi.org/10.46328/ijemst.3761>
- AL-Eidan, A. A., & Almisad, B. M. (2025). Risks of employing generative AI applications (ChatGPT) in education and mitigation approaches: Perspectives from the faculty members at the college of basic education in Kuwait. *Eurasia Journal of Mathematics, Science and Technology Education*, 21(12), em2742. <https://doi.org/10.29333/ejmste/17435>

- Alaj ääski, J. (2006). *How does Web technology affect students' attitudes towards the discipline and study of mathematics/statistics?* *International Journal of Mathematical Education in Science and Technology*, 37*(1), 71–79. <https://doi.org/10.1080/00207390500226002>
- Alazemi, B., & Larkins, R. (2024). Breaking the Math Barrier: Female Students' Perceptions of Math and Their Involvement in a Math Practice Workshop. *Mathematics Teaching-Research Journal*, 16(5), 92–115.
- Alcock, L., Attridge, N., Kenny, S., & Inglis, M. (2014). Achievement and behaviour in undergraduate mathematics: personality is a better predictor than gender. *Research in Mathematics Education*, 16(1), 1–17. <https://doi.org/10.1080/14794802.2013.874094>
- Ali, M., & Tse, A. W. C. (2023). Research Trends and Issues of Engineering Design Process for STEM Education in K-12: A Bibliometric Analysis. *International Journal of Education in Mathematics, Science and Technology*, 11(3), 695–727. <https://doi.org/10.46328/ijemst.2794>
- Ali, R. M., & Kor, L. K. (2007). Association between Brain Hemisphericity, Learning Styles and Confidence in Using Graphics Calculator for Mathematics. *EURASIA Journal of Mathematics, Science and Technology Education*, 3(2). <https://doi.org/10.12973/ejmste/75384>
- Alihodžić, B., Atanasova-Pachemska, T., & Nesimović, S. (2024). Attitudes of high school students towards visualization of mathematical content. *International Electronic Journal of Mathematics Education*, 19(2), em0770. <https://doi.org/10.29333/iejme/14246>
- Alissa, R. A. S., & Hamadne, M. A. (2023). The Level of Science and Mathematics Teachers' Employment of Artificial Intelligence Applications in the Educational Process. *International Journal of Education in Mathematics, Science and Technology*, 11(6), 1597–1608. <https://doi.org/10.46328/ijemst.3806>
- Alkouri, Z., & Wardat, Y. (2025). Digital integration in preschool math: Assessing the efficacy of a structured curriculum in classroom settings. *International Electronic Journal of Mathematics Education*, 20(4), em0845. <https://doi.org/10.29333/iejme/16662>
- Allen, M. M. (2023). High school Black girls' experiences in a STEM after-school program: A qualitative case study. *School Science and Mathematics*, 124(3), 146–158. <https://doi.org/10.1111/ssm.12624>
- Almukhambetova, A., Kuzhabekova, A., & Hernández-Torrano, D. (2022). Hidden Bias, Low Expectations, and Social Stereotypes: Understanding Female Students' Retention in Math-Intensive STEM Fields. *International Journal of Science and Mathematics Education*, 21(2), 535–557. <https://doi.org/10.1007/s10763-022-10256-8>
- Almukhambetova, A., Torrano, D. H., & Nam, A. (2023). Fixing the Leaky Pipeline for Talented Women in STEM. *International Journal of Science and Mathematics Education*, 21(1), 305–324. <https://doi.org/10.1007/s10763-021-10239-1>

- Alrajeh, T. S., & Shindel, B. W. (2020). STUDENT ENGAGEMENT AND MATH TEACHERS SUPPORT. *Journal on Mathematics Education*, 11(2), 167–180.
<https://doi.org/10.22342/jme.11.2.10282.167-180>
- Alsmadi, M. A. (2020). Requirements for Application of the STEM Approach as Perceived by Science, Math and Computer Teachers and their Attitudes towards it. *Eurasia Journal of Mathematics, Science and Technology Education*, 16(9), em1879.
<https://doi.org/10.29333/ejmste/8391>
- Althubaiti, A. (2021). Attitudes of Medical Students Toward Statistics in Medical Research: Evidence From Saudi Arabia. *Journal of Statistics and Data Science Education*, 29(1), 115–121. <https://doi.org/10.1080/10691898.2020.1850220>
- Alzubaidi, . H., Alanazi, . A., & Alwahbi, . T. (2023). The Reality of Applying Educational Electronic Games from Elementary School Teachers' Viewpoint. *International Journal of Education in Mathematics, Science and Technology*, 11(4), 1016–1038.
<https://doi.org/10.46328/ijemst.3526>
- Amador, J. M. (2018). Teachers' gender considerations during elementary mathematics lesson design. *School Science and Mathematics*, 118(7), 290–299.
<https://doi.org/10.1111/ssm.12299>
- Ammann, C., Frauendiener, J., & Holton, D. (2010). German undergraduate mathematics enrolment numbers: background and change. *International Journal of Mathematical Education in Science and Technology*, 41(4), 435–449.
<https://doi.org/10.1080/00207390903564629>
- Amuah, E., & Davis, E. K. (2023). Strategies and Procedural and Conceptual Knowledge of Addition of Unlike Denominator Fractions: The Case of Grade 8 Children in Two Districts of the Central Region of Ghana. *African Journal of Research in Mathematics, Science and Technology Education*, 27(2), 123–136. <https://doi.org/10.1080/18117295.2023.2226546>
- Anasagasti, J., Berciano, A., & Izagirre, A. (2023). A comparison of the effects of different methodologies on the statistics learning profiles of prospective primary education teachers from a gender perspective. *Journal on Mathematics Education*, 14(4), 741–756.
<https://doi.org/10.22342/jme.v14i4.pp741-756>
- Anderson, J. (2002). Gender-related differences on open and closed assessment tasks. *International Journal of Mathematical Education in Science and Technology*, 33(4), 495–503. <https://doi.org/10.1080/00207390210130921>
- Appavoo, P., Armoogum, V., & Soyjaudah, S. (2018). Investigating Mathematics Learning Trajectories: A Comparative Analysis of Grades at Two Major Turning Points. *EURASIA Journal of Mathematics, Science and Technology Education*, 14(4).
<https://doi.org/10.29333/ejmste/82537>
- Arslan, O., & Işıksal-Bostan, M. (2016). Turkish Prospective Middle School Mathematics Teachers' Beliefs and Perceived Self-Efficacy Beliefs Regarding the Use of Origami in

- Mathematics Education. *EURASIA Journal of Mathematics, Science and Technology Education*, 12(6). <https://doi.org/10.12973/eurasia.2016.1243a>
- Asempapa, R. S. (2022). Examining Practicing Teachers' Knowledge and Attitudes toward Mathematical Modeling. *International Journal of Education in Mathematics, Science and Technology*, 10(2), 272–292. <https://doi.org/10.46328/ijemst.2136>
- Atnafu, M. (2014). Secondary School Mathematics Teachers' Attitude in Teaching Mathematics. *International Electronic Journal of Mathematics Education*, 9(1), 57–72. <https://doi.org/10.29333/iejme/281>
- Atweh, B., Bleicher, R. E., & Cooper, T. J. (1998). The Construction of the Social Context of Mathematics Classrooms: A Sociolinguistic Analysis. *Journal for Research in Mathematics Education*, 29(1), 63. <https://doi.org/10.2307/749718>
- Awartani, M., & Gray, M. W. (1989). Cultural influences on sex differentials in mathematics aptitude and achievement. *International Journal of Mathematical Education in Science and Technology*, 20(2), 317–320. <https://doi.org/10.1080/0020739890200212>
- Ayebo, A., & Dingel, M. (2021). Exploring the Relationship among Mathematics Attitude, Gender, and Achievement of Undergraduate Health Science Students. *Mathematics Teaching-Research Journal*, 13(3), 5–20.
- Aytekin, E., & Isiksal-Bostan, M. (2018). Middle school students' attitudes towards the use of technology in mathematics lessons: does gender make a difference? *International Journal of Mathematical Education in Science and Technology*, 50(5), 707–727. <https://doi.org/10.1080/0020739x.2018.1535097>
- Bagaka's, J. G. (2010). THE ROLE OF TEACHER CHARACTERISTICS AND PRACTICES ON UPPER SECONDARY SCHOOL STUDENTS' MATHEMATICS SELF-EFFICACY IN NYANZA PROVINCE OF KENYA: A MULTILEVEL ANALYSIS. *International Journal of Science and Mathematics Education*, 9(4), 817–842. <https://doi.org/10.1007/s10763-010-9226-3>
- Bahi, S., Higgins, D., & Staley, P. (2014). A TIME HAZARD ANALYSIS OF STUDENT PERSISTENCE: A US UNIVERSITY UNDERGRADUATE MATHEMATICS MAJOR EXPERIENCE. *International Journal of Science and Mathematics Education*, 13(5), 1139–1160. <https://doi.org/10.1007/s10763-014-9538-9>
- Bai, Y., Liang, H., Qi, C., & Zuo, S. (2023). An Assessment of Eighth Graders' Mathematics Higher Order Thinking Skills in the Chinese Context. *Canadian Journal of Science, Mathematics and Technology Education*, 23(2), 365–382. <https://doi.org/10.1007/s42330-023-00279-w>
- Bailey, B., Cooper, T. E., & Briggs, K. S. (2012). The Effect of a Modified Moore Method on Attitudes and Beliefs in Precalculus. *School Science and Mathematics*, 112(6), 377–383. <https://doi.org/10.1111/j.1949-8594.2012.00152.x>
- Bakar, N. S. A., Maat, S. M., & Rosli, R. (2020). MATHEMATICS TEACHER'S SELF-EFFICACY OF TECHNOLOGY INTEGRATION AND TECHNOLOGICAL

PEDAGOGICAL CONTENT KNOWLEDGE. *Journal on Mathematics Education*, 11(2), 259–276. <https://doi.org/10.22342/jme.11.2.10818.259-276>

Bansilal, S., & Lephoto, T. (2022). Exploring Particular Learner Factors Associated with South African Mathematics Learners' Achievement: Gender Gap or Not. *African Journal of Research in Mathematics, Science and Technology Education*, 26(2), 77–88. <https://doi.org/10.1080/18117295.2022.2057730>

Battey, D., Amman, K., Leyva, L. A., Hyland, N., & McMichael, E. W. (2022). Racialized and Gendered Labor in Students' Responses to Precalculus and Calculus Instruction. *Journal for Research in Mathematics Education*, 53(2), 94–113. <https://doi.org/10.5951/jresmetheduc-2020-0170>

Bawaneh, A. K., Al-Salman, S. M., & Rababh, B. S. (2023). Does distance teaching of science and mathematics meet teachers' expectations?: Science and mathematics teachers' perceptions, attitudes, and challenges of distance education. *Eurasia Journal of Mathematics, Science and Technology Education*, 19(11), em2358. <https://doi.org/10.29333/ejmste/13810>

Baya'a, N. F. (1990). Mathematics anxiety, mathematics achievement, gender, and socio-economic status among Arab secondary students in Israel. *International Journal of Mathematical Education in Science and Technology*, 21(2), 319–324. <https://doi.org/10.1080/0020739900210221>

Becker, J. R., & Hall, J. (2023). Research on gender and mathematics: exploring new and future directions. *ZDM – Mathematics Education*, 56(1), 141–151. <https://doi.org/10.1007/s11858-023-01510-6>

Beekman, J. A., & Ober, D. (2015). Gender Gap Trends on Mathematics Exams Position Girls and Young Women for STEM Careers. *School Science and Mathematics*, 115(1), 35–50. <https://doi.org/10.1111/ssm.12098>

Berg, D. A. G., Ingram, N., Asil, M., Ward, J., & Smith, J. K. (2024). Self-efficacy in teaching mathematics and the use of effective pedagogical practices in New Zealand primary schools. *Journal of Mathematics Teacher Education*, 28(1), 129–149. <https://doi.org/10.1007/s10857-024-09623-9>

Berry, R. Q. (2008). Access to Upper-Level Mathematics: The Stories of Successful African American Middle School Boys. *Journal for Research in Mathematics Education*, 39(5), 464–488. <https://doi.org/10.5951/jresmetheduc.39.5.0464>

Beswick, K., Watt, H. M. G., Granziera, H., Geiger, V., & Fraser, S. (2023). Boys' motivation profiles in mathematics: relations with contextual factors, wellbeing and engagement in a boys-only school. *ZDM – Mathematics Education*, 55(2), 315–329. <https://doi.org/10.1007/s11858-022-01464-1>

Bicer, A., Capraro, R. M., & Capraro, M. M. (2017). Hispanic students' mathematics achievement in the context of their high school types as STEM and non-STEM schools. *International Journal of Mathematical Education in Science and Technology*, 49(5), 705–720. <https://doi.org/10.1080/0020739x.2017.1410735>

- Billman, A., Harding, A., & Engelbrecht, J. (2018). Does the chalkboard still hold its own against modern technology in teaching mathematics? A case study. *International Journal of Mathematical Education in Science and Technology*, 49(6), 809–823. <https://doi.org/10.1080/0020739x.2018.1431852>
- Birenbaum, M., Nasser, F., & Tatsuoka, C. (2007). Effects of gender and ethnicity on fourth graders' knowledge states in mathematics. *International Journal of Mathematical Education in Science and Technology*, 38(3), 301–319. <https://doi.org/10.1080/00207390601116052>
- Birgin, O., Gürbüz, R., & Memiş, K. Z. (2021). Performance of second-grade elementary school students on counting, place value understanding, and addition operation in natural numbers. *International Journal of Mathematical Education in Science and Technology*, 53(12), 3377–3392. <https://doi.org/10.1080/0020739x.2021.1944680>
- Boaler, J. (1997). Equity, empowerment and different ways of knowing. *Mathematics Education Research Journal*, 9(3), 325–342. <https://doi.org/10.1007/bf03217322>
- Boateng, S., Mudadigwa, B., & Johnston-Wilder, S. (2025). Examining gendered patterns in mathematics and science anxiety levels among physical science pre-service teachers. *Eurasia Journal of Mathematics, Science and Technology Education*, 21(1), em2564. <https://doi.org/10.29333/ejmste/15800>
- Boda, P. A., James, K., Sotelo, J., McGee, S., & Uttal, D. (2021). Racial and gender disparities in elementary mathematics. *School Science and Mathematics*, 122(1), 36–53. <https://doi.org/10.1111/ssm.12506>
- BONDI, C. M. (1987). Girls and Mathematics: A Report of the Joint Mathematical Education Committee of the Royal Society and the Institute of Mathematics and its Applications. *Teaching Mathematics and its Applications*, 6(2), 58–62. <https://doi.org/10.1093/teamat/6.2.58>
- Bondurant, L., Wilburne, J., & Franz, D. (2025). Exploring Teacher Candidates' Responses to Equity Vignettes. *Journal of Urban Mathematics Education*, 18(2), 19–49. <https://doi.org/10.21423/jume-v18i2a743>
- Bordás, A., & Debrenti, E. (2025). University students' logical thinking in the light of success and perceived difficulty. *International Journal of Mathematical Education in Science and Technology*, 56(8), 1554–1566. <https://doi.org/10.1080/0020739x.2025.2514102>
- Brandell, G. (2008). Progress and stagnation of gender equity: contradictory trends within mathematics research and education in Sweden. *ZDM*, 40(4), 659–672. <https://doi.org/10.1007/s11858-008-0127-7>
- Brandell, G., Carlsson, S., Ekblom, H., & Nord, A. C. (1997). Encouraging more women into computer science: Initiating a single-sex intervention program in Sweden. *Mathematics Education Research Journal*, 9(3), 312–324. <https://doi.org/10.1007/bf03217321>

- Brandell, G., Leder, G., & Nyström, P. (2007). Gender and Mathematics: recent development from a Swedish perspective. *ZDM*, 39(3), 235–250. <https://doi.org/10.1007/s11858-007-0025-4>
- Bremigan, E. G. (2005). An Analysis of Diagram Modification and Construction in Students' Solutions to Applied Calculus Problems. *Journal for Research in Mathematics Education*, 36(3), 248–277.
- Brignell, C., Noyes, A., & Jacques, L. (2024). The mathematics pipeline in England: inclusion and the excellence stream. *Teaching Mathematics and its Applications: An International Journal of the IMA*, 44(1), 28–46. <https://doi.org/10.1093/teamat/hrae005>
- Brnic, M., Greefrath, G., & Reinhold, F. (2024). Working with digital textbooks or printed materials: A study with boys and girls on conditional probability. *ZDM – Mathematics Education*, 56(4), 559–572. <https://doi.org/10.1007/s11858-023-01543-x>
- Brown, L. I., & Kanyongo, G. Y. (2010). Gender Differences in Mathematics Performance in Trinidad and Tobago: Examining Affective Factors. *International Electronic Journal of Mathematics Education*, 5(3), 113–130. <https://doi.org/10.29333/iejme/253>
- Brown, M., Brown, P., & Bibby, T. (2008). “I would rather die”: reasons given by 16-year-olds for not continuing their study of mathematics. *Research in Mathematics Education*, 10(1), 3–18. <https://doi.org/10.1080/14794800801915814>
- Büchele, S., & Feiste, L. (2024). Exploring determinants of engagement with voluntary mathematics assignments in economics education: a correlational study. *Teaching Mathematics and its Applications: An International Journal of the IMA*, 43(4), 339–355. <https://doi.org/10.1093/teamat/hrae016>
- Büchele, S., Liebendörfer, M., & Lankeit, E. (2021). Increasing the effect of a remedial mathematics course by switching to an online format during the COVID-19 crisis: evidence from a German university. *Teaching Mathematics and its Applications: An International Journal of the IMA*, 40(4), 478–496. <https://doi.org/10.1093/teamat/hrab013>
- BULUT, A. S., YILDIZ, A., & BALTACI, S. (2020). A Comparison of Mathematics Learning Approaches of Gifted and Non-Gifted Students. *Turkish Journal of Computer and Mathematics Education (TURCOMAT)*. <https://doi.org/10.16949/turkbilmat.682111>
- Bursal, M. (2009). TURKISH PRESERVICE ELEMENTARY TEACHERS' SELF-EFFICACY BELIEFS REGARDING MATHEMATICS AND SCIENCE TEACHING. *International Journal of Science and Mathematics Education*, 8(4), 649–666. <https://doi.org/10.1007/s10763-009-9179-6>
- Burton†, L. (2008). Moving towards a feminist epistemology of mathematics. *ZDM*, 40(4), 519–528. <https://doi.org/10.1007/s11858-008-0109-9>
- Busaka, C., Umugiraneza, O., & Kitta, S. R. (2022). Mathematics teachers' conceptual understanding of soft skills in secondary schools in Zambia. *Eurasia Journal of Mathematics, Science and Technology Education*, 18(7), em2128. <https://doi.org/10.29333/ejmste/12160>

- Butakor, P. K., Ampadu, E., & Cole, Y. (2017). Ghanaian Students in TIMSS 2011: Relationship between Contextual Factors and Mathematics Performance. *African Journal of Research in Mathematics, Science and Technology Education*, 21(3), 316–326. <https://doi.org/10.1080/18117295.2017.1379281>
- Cambria, J., Brandt, H., Nagengast, B., & Trautwein, U. (2017). Frame of Reference effects on values in mathematics: evidence from German secondary school students. *ZDM*, 49(3), 435–447. <https://doi.org/10.1007/s11858-017-0841-0>
- Campbell, T., Miles, S., & Kularajan, S. (2024). Geographical and methodological trends in ESM and JRME from 2014–2023. *Investigations in Mathematics Learning*, 17(4), 346–365. <https://doi.org/10.1080/19477503.2024.2439212>
- Carr, M., Jessup, D. L., & Fuller, D. (1999). Gender Differences in First-Grade Mathematics Strategy Use: Parent and Teacher Contributions. *Journal for Research in Mathematics Education*, 30(1), 20. <https://doi.org/10.2307/749628>
- Casey, M. B., Nuttall, R. L., & Pezaris, E. (2001). Spatial-Mechanical Reasoning Skills versus Mathematics Self-Confidence as Mediators of Gender Differences on Mathematics Subtests Using Cross-National Gender-Based Items. *Journal for Research in Mathematics Education*, 32(1), 28. <https://doi.org/10.2307/749620>
- Caspi, A., & Gorsky, P. (2025). Students' Motivation to Learn Mathematics Predicts their STEM Career Expectations and Mediates the Effects of Gender, Parental Occupations and Math Classroom Environments. *International Journal of Science and Mathematics Education*, 23(8), 3627–3658. <https://doi.org/10.1007/s10763-025-10581-8>
- Castaneda, A., Aguilar, M. S., Gómez-Blancarte, A. L., Romo-Vázquez, A., Lezama-Andalón, J., & Miranda-Viramontes, I. (2019). Representations of Mathematicians in Lower Secondary Mathematics Textbooks. *EURASIA Journal of Mathematics, Science and Technology Education*, 15(7). <https://doi.org/10.29333/ejmste/106056>
- Castillo, A. J., Durán, P., Fuller, E., Watson, C., Potvin, G., & Kramer, L. H. (2024). Student attitudes in an innovative active learning approach in calculus. *International Journal of Mathematical Education in Science and Technology*, 55(3), 730–758. <https://doi.org/10.1080/0020739x.2021.2023771>
- Çatlıoğlu, H., Gürbüz, R., & Birgin, O. (2014). Do pre-service elementary school teachers still have mathematics anxiety? Some factors and correlates. *Bolema: Boletim de Educação Matemática*, 28(48), 110–127. <https://doi.org/10.1590/1980-4415v28n48a06>
- Çelik, H. C. (2020). Development of symbolic mathematics language literacy perception scale for junior high school students and analysis of literacy perceptions based on certain variables. *International Journal of Mathematical Education in Science and Technology*, 51(3), 327–343. <https://doi.org/10.1080/0020739x.2018.1555724>
- Chacon, P., & Soto-Johnson, H. (2003). Encouraging Young Women to Stay in the Mathematics Pipeline: Mathematics Camps for Young Women. *School Science and Mathematics*, 103(6), 274–284. <https://doi.org/10.1111/j.1949-8594.2003.tb18150.x>

- Chang, P. (1986). Academic performance, sex difference and attitudes toward mathematics of small groups as factors in college developmental mathematics courses. *International Journal of Mathematical Education in Science and Technology*, 17(2), 157–168. <https://doi.org/10.1080/0020739860170205>
- Charitaki, G., Vretudaki, H., & Kypriotaki, M. (2025). Assessing mathematics teaching efficacy profiles in kindergarten: The impact of special educational needs, school climate, responsibility for student achievement, self-efficacy, and locus of control. *Asian Journal for Mathematics Education*, 4(3), 394–415. <https://doi.org/10.1177/27527263251329953>
- Cheema, J. R., & Kitsantas, A. (2013). INFLUENCES OF DISCIPLINARY CLASSROOM CLIMATE ON HIGH SCHOOL STUDENT SELF-EFFICACY AND MATHEMATICS ACHIEVEMENT: A LOOK AT GENDER AND RACIAL–ETHNIC DIFFERENCES. *International Journal of Science and Mathematics Education*, 12(5), 1261–1279. <https://doi.org/10.1007/s10763-013-9454-4>
- Chiang, P. J., Lin, Y. W., & Tseng, C. L. (2016). The Effect of Blended Learning in Mathematics Course. *EURASIA Journal of Mathematics, Science and Technology Education*, 13(3). <https://doi.org/10.12973/eurasia.2017.00641a>
- Chikunda, C. (2014). Identifying Tensions around Gender-responsive Curriculum Practices in Science Teacher Education in Zimbabwe: An Activity Theory Analysis. *African Journal of Research in Mathematics, Science and Technology Education*, 18(3), 264–275. <https://doi.org/10.1080/10288457.2014.956409>
- Chirume, S. (2017). Learning Mathematics: Like Proposing to a Girl. *EURASIA Journal of Mathematics, Science and Technology Education*, 14(2). <https://doi.org/10.12973/ejmste/76962>
- Chiu, M. S. (2018). Effects of Early Numeracy Activities on Mathematics Achievement and Affect: Parental Value and Child Gender Conditions and Socioeconomic Status Mediation. *EURASIA Journal of Mathematics, Science and Technology Education*, 14(12). <https://doi.org/10.29333/ejmste/97191>
- Chodnowsky, A., & Columba, L. (2018). Methods of mitigating the effects of stereotype threat on female students' mathematics performance in a high school geometry classroom. *Mathematics Teaching-Research Journal*, 10(2), 10–28.
- Christensen, R., & Knezek, G. (2020). Indicators of middle school students' mathematics enjoyment and confidence. *School Science and Mathematics*, 120(8), 491–503. <https://doi.org/10.1111/ssm.12439>
- Chronaki, A., & Kollosche, D. (2019). Refusing mathematics: a discourse theory approach on the politics of identity work. *ZDM*, 51(3), 457–468. <https://doi.org/10.1007/s11858-019-01028-w>
- Chu, P. K. K. (2024). Attitudes toward statistics among business students: Do majors, mathematical background, sex, and field of secondary education matter? *Teaching Statistics*, 47(1), 63–74. <https://doi.org/10.1111/test.12390>

- ÇİFTÇİ, S. K., & YILDIZ, P. (2019). The Effect of Gender on Algebra Achievement: The Meta-Analysis of Trends in International Mathematics and Science Study (TIMSS). *Turkish Journal of Computer and Mathematics Education (TURCOMAT)*, 10(3), 617–627. <https://doi.org/10.16949/turkbilmater.568545>
- Cooper, T. E., Bailey, B., & Briggs, K. S. (2015). Gender Differences in Achievement in an Inquiry-Based Learning Precalculus Course. *International Electronic Journal of Mathematics Education*, 10(2), 97–110. <https://doi.org/10.29333/iejme/294>
- Copur-Gencturk, Y., Thacker, I., & Quinn, D. (2020). K-8 Teachers' Overall and Gender-Specific Beliefs About Mathematical Aptitude. *International Journal of Science and Mathematics Education*, 19(6), 1251–1269. <https://doi.org/10.1007/s10763-020-10104-7>
- Corrigan, E., Williams, M., & Wells, M. A. (2023). High School Enrolment Choices—Understanding the STEM Gender Gap. *Canadian Journal of Science, Mathematics and Technology Education*, 23(3), 403–421. <https://doi.org/10.1007/s42330-023-00285-y>
- Cribbs, J. D., & Piatek-Jimenez, K. (2022). Exploring How Gender, Self-Identified Personality Attributes, Mathematics Identity, and Gender Identification Contribute to College Students' STEM Career Goals. *International Journal of Innovation in Science and Mathematics Education*, 29(2). <https://doi.org/10.30722/ijisme.29.02.004>
- da Silva, H. A., & Moura, A. S. (2020). Teaching Introductory Statistical Classes in Medical Schools Using RStudio and R Statistical Language: Evaluating Technology Acceptance and Change in Attitude Toward Statistics. *Journal of Statistics Education*, 28(2), 212–219. <https://doi.org/10.1080/10691898.2020.1773354>
- Damarin, S., & Erchick, D. B. (2010). Research Commentary: Toward Clarifying the Meanings of Gender in Mathematics Education Research. *Journal for Research in Mathematics Education*, 41(4), 310–323. <https://doi.org/10.5951/jresmetheduc.41.4.0310>
- Darragh, L. (2013). Constructing confidence and identities of belonging in mathematics at the transition to secondary school. *Research in Mathematics Education*, 15(3), 215–229. <https://doi.org/10.1080/14794802.2013.803775>
- Darragh, L. (2018). Loving and Loathing: Portrayals of School Mathematics in Young Adult Fiction. *Journal for Research in Mathematics Education*, 49(2), 178–209. <https://doi.org/10.5951/jresmetheduc.49.2.0178>
- Davis, J. D., & Shih, J. C. (2007). Secondary Options and Post-secondary Expectations: Standards-based Mathematics Programs and Student Achievement on College Mathematics Placement Exams. *School Science and Mathematics*, 107(8), 336–346. <https://doi.org/10.1111/j.1949-8594.2007.tb17798.x>
- Dawson, C. (2019). Tackling limited spatial ability: lowering one barrier into STEM? *European Journal of Science and Mathematics Education*, 7(1), 14–31. <https://doi.org/10.30935/scimath/9531>
- De Loof, H., Boeve-de Pauw, J., & Van Petegem, P. (2021). Engaging Students with Integrated STEM Education: a Happy Marriage or a Failed Engagement? *International Journal of*

Science and Mathematics Education, 20(7), 1291–1313. <https://doi.org/10.1007/s10763-021-10159-0>

Deakin, M. A. B. (2012). Theano: the world's first female mathematician? *International Journal of Mathematical Education in Science and Technology*, 44(3), 350–364. <https://doi.org/10.1080/0020739x.2012.729614>

Del Olmo-Muñoz, J., Diago, P. D., Arnau, D., Arnau-Blasco, D., & González-Calero, J. A. (2024). Metacognitive control during problem solving at early ages in programming tasks using a floor robot. *ZDM – Mathematics Education*, 56(6), 1303–1317. <https://doi.org/10.1007/s11858-024-01621-8>

Desbiens, V., Larose, S., & Boisclair Châteauvert, G. (2023). Animations scientifiques en classe de primaire: des effets motivationnels pour qui et pour quelle intensité de participation? *Canadian Journal of Science, Mathematics and Technology Education*, 23(4), 664–686. <https://doi.org/10.1007/s42330-024-00310-8>

Dindyal, J. (2008). An overview of the gender factor in mathematics in TIMSS-2003 for the Asia-Pacific region. *ZDM*, 40(6), 993–1005. <https://doi.org/10.1007/s11858-008-0111-2>

Dintarini, M., Budiarto, M. T., & Fuad, Y. (2025). Spatial Thinking in Conic Sections: A Study of Undergraduate Mathematics Students by Sex and Spatial Category. *Jurnal Pendidikan Matematika*, 19(2), 275–298. <https://doi.org/10.22342/mej.v19i2.pp275-298>

Dogucu, M., Johnson, A. A., & Ott, M. (2023). Framework for Accessible and Inclusive Teaching Materials for Statistics and Data Science Courses. *Journal of Statistics and Data Science Education*, 31(2), 144–150. <https://doi.org/10.1080/26939169.2023.2165988>

Dominguez, H., Takeuchi, M. A., & Civil, M. (2023). Three embodied voices speaking on/to research on language, mathematics, and the learner. *ZDM – Mathematics Education*, 55(6), 1037–1051. <https://doi.org/10.1007/s11858-023-01525-z>

Dorce, C. (2025). History of mathematics (and mathematics) with gender perspective: towards a social history of mathematics. *International Journal of Mathematical Education in Science and Technology*, 56(3), 363–381. <https://doi.org/10.1080/0020739x.2023.2236136>

Doz, D., & Sliško, J. (2024). The cognitive reflection test and students' achievements in mathematics and physics. *European Journal of Science and Mathematics Education*, 12(1), 85–96. <https://doi.org/10.30935/scimath/13832>

Duchhardt, C., Jordan, A. K., & Ehmke, T. (2017). Adults' Use of Mathematics and Its Influence on Mathematical Competence. *International Journal of Science and Mathematics Education*, 15(1), 155–174. <https://doi.org/10.1007/s10763-015-9670-1>

Duzenli Gokalp, N. (2020). Mathematics – How do students and teachers perceive it? *International Journal of Innovation in Science and Mathematics Education*, 28(1). <https://doi.org/10.30722/ijisme.28.01.002>

Edosomwan, K., & Williams, J. A. (2023). Double jeopardy? Examining the influence of mathematics tracking on in-school suspensions through an intersectionality framework. *School Science and Mathematics*, 124(2), 72–84. <https://doi.org/10.1111/ssm.12607>

- Egara, F. O., & Mosimege, M. D. (2023). Gender difference in secondary school students' retention in algebra: A computer simulation approach. *Eurasia Journal of Mathematics, Science and Technology Education*, 19(7), em2290. <https://doi.org/10.29333/ejmste/13280>
- Eilam, B., & Alon, U. (2018). Children's Object Structure Perspective-Taking: Training and Assessment. *International Journal of Science and Mathematics Education*, 17(8), 1541–1562. <https://doi.org/10.1007/s10763-018-9934-7>
- Ejembi, Oche, S., Oguguo, B. C. E., Ezeanya, C. O., Okpe, K. E., Okwara, V. C., Adie, I. I., Ayek, B. B., Okpara, L. I., & Ojobo, B. N. (2024). Relative Effectiveness of Formative Assessment Techniques on Students' Academic Achievement in Mathematics Classroom Teaching and Learning. *Mathematics Teaching-Research Journal*, 15(6), 171–193.
- El Chaabi, M., & Younes, R. (2025). The effect of digital storytelling on middle school students' interests in STEM fields and stereotyping. *Eurasia Journal of Mathematics, Science and Technology Education*, 21(4), em2621. <https://doi.org/10.29333/ejmste/16220>
- Ellis, B., & Elizabeth Wrightsman (2024). An Anti-deficit Counter-story of a Black Girl's Forms of Resilience in a Standards-based Mathematics Classroom. *Journal of Urban Mathematics Education*, 17(1), 48–83. <https://doi.org/10.21423/jume-v17i1a514>
- Er, Z., & Dinç Artut, P. (2023). Metaphorical perceptions of gifted and normally developing students on the concept of solving mathematical problems. *Journal on Mathematics Education*, 14(4), 627–646. <https://doi.org/10.22342/jme.v14i4.pp627-646>
- Ercikan, K., McCreith, T., & Lapointe, V. (2005). Factors Associated With Mathematics Achievement and Participation in Advanced Mathematics Courses: An Examination of Gender Differences From an International Perspective. *School Science and Mathematics*, 105(1), 5–14. <https://doi.org/10.1111/j.1949-8594.2005.tb18031.x>
- Ernest, P. (1995). Values, gender and images of mathematics: a philosophical perspective. *International Journal of Mathematical Education in Science and Technology*, 26(3), 449–462. <https://doi.org/10.1080/0020739950260313>
- Eshetu, F., & Assefa, S. (2018). Effects of Context-Based Instructional Approaches on Students' Problem-Solving Skills in Rotational Motion. *Eurasia Journal of Mathematics Science and Technology Education*, 15(2). <https://doi.org/10.29333/ejmste/102283>
- Esmonde, I., & Langer-Osuna, J. M. (2013). Power in Numbers: Student Participation in Mathematical Discussions in Heterogeneous Spaces. *Journal for Research in Mathematics Education*, 44(1), 288–315. <https://doi.org/10.5951/jresmetheduc.44.1.0288>
- ESNARD, T. R., ALLADIN, F. M., & SAMLAL, K. C. (2021). PRIOR MATHEMATICS PERFORMANCE, STATISTICS ANXIETY, SELF-EFFICACY AND EXPECTATIONS FOR PERFORMANCE IN STATISTICS: A SURVEY OF SOCIAL SCIENCES STUDENTS IN A CARIBBEAN INSTITUTION OF HIGHER EDUCATION. *STATISTICS EDUCATION RESEARCH JOURNAL*, 20(1), 4. <https://doi.org/10.52041/serj.v20i1.98>

- Esparza Puga, D. S., & Sánchez Aguilar, M. (2025). Exploring the role of physical appearance in students' evaluations of female mathematics teachers' competencies. *ZDM – Mathematics Education*, 57(7), 1383–1396. <https://doi.org/10.1007/s11858-025-01739-3>
- Esperanza, P. J., Himang, C., Bongo, M., Selerio Jr., E., & Ocampo, L. (2023). The utility of a flipped classroom in secondary Mathematics education. *International Journal of Mathematical Education in Science and Technology*, 54(3), 382–415. <https://doi.org/10.1080/0020739x.2021.1957166>
- Evans, B. R. (2013). Beliefs about social justice among elementary mathematics teachers. *Mathematics Teaching-Research Journal*, 6(1-2), 46–58.
- Fan, L., Luo, J., Xie, S., Zhu, F., & Li, S. (2022). Chinese students' access, use and perceptions of ICTs in learning mathematics: findings from an investigation of Shanghai secondary schools. *ZDM – Mathematics Education*, 54(3), 611–624. <https://doi.org/10.1007/s11858-022-01363-5>
- Fan, L., Xie, S., Luo, J., Li, L., Tang, J., & Li, S. (2023). Teachers' perceptions of less successfully organized professional development practices in mathematics: A study of nine secondary schools in Shanghai, China. *Journal of Mathematics Teacher Education*, 26(5), 667–697. <https://doi.org/10.1007/s10857-023-09591-6>
- Faradillah, A., & Rizkia, P. A. (2024). Mathematical Understanding Concepts Ability in QR Code-Assisted Problem Solving by Gender. *Jurnal Pendidikan Matematika*, 18(3), 313–328. <https://doi.org/10.22342/jpm.v18i3.pp313-328>
- Fayowski, V., & MacMillan, P. D. (2008). An evaluation of the Supplemental Instruction programme in a first year calculus course. *International Journal of Mathematical Education in Science and Technology*, 39(7), 843–855. <https://doi.org/10.1080/00207390802054433>
- Fayowski, V., Hyndman, J., & MacMillan, P. D. (2009). Assessment on Previous Course Work in Calculus and Subsequent Achievement in Calculus at the Post-Secondary Level. *Canadian Journal of Science, Mathematics and Technology Education*, 9(1), 49–57. <https://doi.org/10.1080/14926150902853204>
- Fernandez, F., Froschl, M., Lorenzetti, L., & Stimmer, M. (2024). Investigating the importance of girls' mathematical identity within United States STEM programmes: a systematic review. *International Journal of Mathematical Education in Science and Technology*, 55(3), 650–690. <https://doi.org/10.1080/0020739x.2021.2022229>
- Ferretti, F., & Giberti, C. (2020). The Properties of Powers: Didactic Contract and Gender Gap. *International Journal of Science and Mathematics Education*, 19(8), 1717–1735. <https://doi.org/10.1007/s10763-020-10130-5>
- Fleener, M. J. (1996). Scientific World Building on the Edge of Chaos: High School Students' Beliefs About Mathematics and Science. *School Science and Mathematics*, 96(6), 312–320. <https://doi.org/10.1111/j.1949-8594.1996.tb15841.x>

- Forgasz, H. (2010). Streaming for mathematics in years 7–10 in Victoria: An issue of equity? *Mathematics Education Research Journal*, 22(1), 57–90.
<https://doi.org/10.1007/bf03217559>
- Forgasz, H. (2021). Gender: A dilemma for large-scale studies in mathematics education. *Mathematics Education Research Journal*, 33(4), 631–640.
<https://doi.org/10.1007/s13394-020-00353-8>
- Forgasz, H. J., & Hill, J. C. (2012). FACTORS IMPLICATED IN HIGH MATHEMATICS ACHIEVEMENT. *International Journal of Science and Mathematics Education*, 11(2), 481–499. <https://doi.org/10.1007/s10763-012-9348-x>
- Forgasz, H. J., & Leder, G. C. (1996). Mathematics classrooms, gender and affect. *Mathematics Education Research Journal*, 8(2), 153–173. <https://doi.org/10.1007/bf03217295>
- Forgasz, H. J., & Leder, G. C. (2017). Persistent gender inequities in mathematics achievement and expectations in Australia, Canada and the UK. *Mathematics Education Research Journal*, 29(3), 261–282. <https://doi.org/10.1007/s13394-017-0190-x>
- Forgasz, H. J., & Mittelberg, D. (2008). Israeli Jewish and Arab students' gendering of mathematics. *ZDM*, 40(4), 545–558. <https://doi.org/10.1007/s11858-008-0139-3>
- Forgasz, H. J., Leder, G. C., & Gardner, P. L. (1999). The Fennema-Sherman Mathematics as a Male Domain Scale Reexamined. *Journal for Research in Mathematics Education*, 30(3), 342. <https://doi.org/10.2307/749839>
- Forgasz, H., & Leder, G. (2019). VCE STEM subject enrolments in co-educational and single-sex schools. *Mathematics Education Research Journal*, 32(3), 433–448.
<https://doi.org/10.1007/s13394-019-00271-4>
- Forgasz, H., & Leder, G. (2022). The NAPLAN numeracy test: do school type and socio-economic background make a difference? *Mathematics Education Research Journal*, 34(1), 1–14. <https://doi.org/10.1007/s13394-020-00326-x>
- Forster, P. A., & Mueller, U. (2001). Outcomes and implications of students' use of graphics calculators in the public examination of calculus. *International Journal of Mathematical Education in Science and Technology*, 32(1), 37–52.
<https://doi.org/10.1080/00207390116972>
- Fortune, A., & Spangenberg, E. D. (2023). Trait- and State-Components of Mathematics Anxiety versus Perceived Mathematics Anxiety. *International Journal of Education in Mathematics, Science and Technology*, 11(6), 1366–1385.
<https://doi.org/10.46328/ijemst.2958>
- Franz-Odendaal, T. A., Blotnicky, K. A., & Joy, P. (2020). Math Self-efficacy and the Likelihood of Pursuing a STEM-Based Career: a Gender-Based Analysis. *Canadian Journal of Science, Mathematics and Technology Education*, 20(3), 538–556.
<https://doi.org/10.1007/s42330-020-00105-7>

- Freedman, E. S. (1985). Gender differences in basic numeracy and mathematical development. *International Journal of Mathematical Education in Science and Technology*, 16(4), 507–513. <https://doi.org/10.1080/0020739850160406>
- Furinghetti, F. (2008). The emergence of women on the international stage of mathematics education. *ZDM*, 40(4), 529–543. <https://doi.org/10.1007/s11858-008-0131-y>
- Gallant, C., Bork, P., Carpenter-Cleland, C., & Good, D. (2020). Examining the Impact of a 2-Day Scientific Conference on High School Students' Interest in STEM and Confidence in Attending University. *Canadian Journal of Science, Mathematics and Technology Education*, 20(2), 376–387. <https://doi.org/10.1007/s42330-020-00086-7>
- Gao, H., Evans, T., & Fergusson, A. (2025). Student-generated explanation in undergraduate mathematics and statistics education: a systematic literature review. *International Journal of Mathematical Education in Science and Technology*, 56(12), 2534–2557. <https://doi.org/10.1080/0020739x.2025.2556867>
- Gao, J. (2019). Sources of Mathematics Self-Efficacy in Chinese Students: a Mixed-Method Study with Q-Sorting Procedure. *International Journal of Science and Mathematics Education*, 18(4), 713–732. <https://doi.org/10.1007/s10763-019-09984-1>
- Garba, A., Ismail, N., Osman, S., & Rameli, M. R. M. (2019). Exploring Peer Effect on Mathematics Anxiety among Secondary School Students of Sokoto State, Nigeria through Photovoice Approach. *EURASIA Journal of Mathematics, Science and Technology Education*, 16(2). <https://doi.org/10.29333/ejmste/112622>
- Gargroetzi, E. C. (2024). Identity, Power, and Dignity: A Positional Analysis of Gisela in Her High School Mathematics Classroom. *Journal for Research in Mathematics Education*, 55(3), 127–147. <https://doi.org/10.5951/jresematheduc-2022-0027>
- Gawlick, T. (2002). On Dynamic Geometry Software in the regular classroom. *Zentralblatt für Didaktik der Mathematik*, 34(3), 85–92. <https://doi.org/10.1007/bf02655711>
- George, L. (2020). Exploring the M in STEM: Post-secondary Participation, Performance and Attrition in Mathematics. *Canadian Journal of Science, Mathematics and Technology Education*, 20(3), 441–461. <https://doi.org/10.1007/s42330-020-00095-6>
- Getenet, S. (2023). The influence of students' prior numeracy achievement on later numeracy achievement as a function of gender and year levels. *Mathematics Education Research Journal*, 36(4), 745–766. <https://doi.org/10.1007/s13394-023-00469-7>
- Gholami, H. (2023). Performance of Malaysian Foundation Level Students in Mathematical Problem Solving As Well As Gender Comparison. *Mathematics Teaching-Research Journal*, 15(2), 104–120.
- Gholson, M. L., & Martin, D. B. (2019). Blackgirl face: racialized and gendered performativity in mathematical contexts. *ZDM*, 51(3), 391–404. <https://doi.org/10.1007/s11858-019-01051-x>
- Giannakopoulou, E. (2023). Reading and understanding numerical information about COVID-19 by Greek adults. *Adults Learning Mathematics International Journal*, 17(1), 7–25.

- Gjøvik, Ø., Kaspersen, E., & Farsani, D. (2022). Stereotypical images of male and female mathematics teachers. *Research in Mathematics Education*, 25(2), 178–193. <https://doi.org/10.1080/14794802.2022.2041471>
- Godfrey, R., & Aubrey, C. (2001). ASSESSING EARLY MATHEMATICAL DEVELOPMENT. *Research in Mathematics Education*, 3(1), 55–68. <https://doi.org/10.1080/14794800008520084>
- Gokhool, F., Lawson, D., Hodds, M., & Aslam, F. (2021). Exploring differential engagement with mathematics support from an engineering student focus. *Teaching Mathematics and its Applications: An International Journal of the IMA*, 41(4), 292–304. <https://doi.org/10.1093/teamat/hrab033>
- Gonzalez-DeHass, A. R., Furner, J. M., Vásquez-Colina, M. D., & Morris, J. D. (2023). Undergraduate Students' Math Anxiety: the Role of Mindset, Achievement Goals, and Parents. *International Journal of Science and Mathematics Education*, 22(5), 1037–1056. <https://doi.org/10.1007/s10763-023-10416-4>
- Goodchild, S., & Grevholm, B. (2009). An Exploratory Study of Mathematics Test Results: What is the Gender Effect? *International Journal of Science and Mathematics Education*, 7(1), 161–182. <https://doi.org/10.1007/s10763-007-9114-7>
- Goonatilake, R., Lewis, K. D., Lin, R., & Kidd, C. E. (2016). Improving gender disparity in scholarship programs for secondary-level mathematics teachers. *Mathematics Teaching-Research Journal*, 8(1-2).
- Grasset, N. (1991). Why girls don't count. *Mathematics Education Research Journal*, 3(2), 1–13. <https://doi.org/10.1007/bf03217223>
- Green, E., Rupnow, T. J., & Bracewell, L. N. (2024). Creating a space for them through autoethnography. *Journal of Mathematics Teacher Education*, 28(7), 1637–1658. <https://doi.org/10.1007/s10857-024-09643-5>
- Grootenboer, P., & Hemmings, B. (2007). Mathematics performance and the role played by affective and background factors peter grootenboer and brian hemmings. *Mathematics Education Research Journal*, 19(3), 3–20. <https://doi.org/10.1007/bf03217459>
- Guerrero, S., Atherton, A., Rushall, A., & Daugherty, R. (2020). Does a Modified Math Emporium Work for All Students? *International Journal of Education in Mathematics, Science and Technology*, 8(3), 233. <https://doi.org/10.46328/ijemst.v8i3.871>
- Guppy, N., Balzarini, M., Kolpashnikova, K., & Lyon, K. (2025). Revisiting the STEM Acronym: Toward Conceptual Clarity. *Canadian Journal of Science, Mathematics and Technology Education*, 25(2), 463–474. <https://doi.org/10.1007/s42330-025-00375-z>
- Gur, T., Balta, N., Dauletkulova, A., Assanbayeva, G., & Fernández-César, R. (2023). Mathematics achievement emotions of high school students in Kazakhstan. *Journal on Mathematics Education*, 14(3), 525–544. <https://doi.org/10.22342/jme.v14i3.pp525-544>

- Gutiérrez, R., Kokka, K., & Myers, M. (2024). Political Conocimiento in Teaching Mathematics: mathematics teacher candidates enacting their ethical identities. *Journal of Mathematics Teacher Education*, 27(5), 755–781. <https://doi.org/10.1007/s10857-024-09627-5>
- Haciomeroglu, E. S., & Chicken, E. (2012). Visual thinking and gender differences in high school calculus. *International Journal of Mathematical Education in Science and Technology*, 43(3), 303–313. <https://doi.org/10.1080/0020739x.2011.618550>
- Haciomeroglu, E. S., Chicken, E., & Dixon, J. K. (2013). Relationships between Gender, Cognitive Ability, Preference, and Calculus Performance. *Mathematical Thinking and Learning*, 15(3), 175–189. <https://doi.org/10.1080/10986065.2013.794255>
- Haley, M. R., Johnson, M. F., & Kuennen, E. W. (2007). Student and Professor Gender Effects in Introductory Business Statistics. *Journal of Statistics Education*, 15(3). <https://doi.org/10.1080/10691898.2007.11889549>
- Halimah, A., & Azina, I. N. (2012). Student Factors and Mathematics Achievement: Evidence from TIMSS 2007. *EURASIA Journal of Mathematics, Science and Technology Education*, 8(4). <https://doi.org/10.12973/eurasia.2012.843a>
- Hall, J., & Suurtamm, C. (2018). Behind the “Success Story”: Exploring the Experiences of a Woman Mathematics Major. *Canadian Journal of Science, Mathematics and Technology Education*, 18(4), 342–354. <https://doi.org/10.1007/s42330-018-0032-1>
- Hall, J., Robinson, T., Flegg, J., & Wilkinson, J. (2022). First-year and final-year undergraduate students’ perceptions of university mathematics departments. *Mathematics Education Research Journal*, 34(2), 189–214. <https://doi.org/10.1007/s13394-020-00340-z>
- Han, S., Samaroo, D., Liou-Mark, J., Kostadinov, B., Thiel, J., & Zeng, S. (2025). Professional Development Workshops for Faculty on Addressing Implicit Bias in Computer Science Education. *Mathematics Teaching-Research Journal*, 17(4), 80–97.
- Hanna, G. (2000). Declining gender differences from FIMS to TIMSS. *Zentralblatt für Didaktik der Mathematik*, 32(1), 11–17. <https://doi.org/10.1007/bf02652734>
- Hannan, S., Juan, A., & Harvey, J. (2024). Fostering Self-efficacy: Understanding South African Grade 9 Students’ Confidence in their Mathematics Abilities. *African Journal of Research in Mathematics, Science and Technology Education*, 28(2), 272–284. <https://doi.org/10.1080/18117295.2024.2404344>
- Hannula, M. S. (2012). Exploring new dimensions of mathematics-related affect: embodied and social theories. *Research in Mathematics Education*, 14(2), 137–161. <https://doi.org/10.1080/14794802.2012.694281>
- Hao, L., Liang, H., Qi, C., & Jiang, C. (2024). Assessing Mathematical Deductive Reasoning Competence of Eighth-Grade Students from China. *Canadian Journal of Science, Mathematics and Technology Education*, 24(1), 84–106. <https://doi.org/10.1007/s42330-024-00315-3>
- Haroun, R. F., Ng, D., Abdelfattah, F. A., & AlSalouli, M. S. (2015). Gender Difference in Teachers’ Mathematical Knowledge for Teaching in the Context of Single-Sex

- Classrooms. *International Journal of Science and Mathematics Education*, 14(S2), 383–396. <https://doi.org/10.1007/s10763-015-9631-8>
- Hasenhütl, S., Luttenberger, S., Macher, D., Eichen, L., Eglmaier, M. T. W., & Paechter, M. (2024). Empowering educators: A training for pre-service and in-service teachers on gender-sensitive STEM instruction. *Eurasia Journal of Mathematics, Science and Technology Education*, 20(6), em2452. <https://doi.org/10.29333/ejmste/14590>
- Healy, L., & Hoyles, C. (2000). A Study of Proof Conceptions in Algebra. *Journal for Research in Mathematics Education*, 31(4), 396. <https://doi.org/10.2307/749651>
- HEDGES, S. (2017). STATISTICS STUDENT PERFORMANCE AND ANXIETY: COMPARISONS IN COURSE DELIVERY AND STUDENT CHARACTERISTICS. *STATISTICS EDUCATION RESEARCH JOURNAL*, 16(1), 320–336. <https://doi.org/10.52041/serj.v16i1.233>
- Heidari, R., & Rajabi, F. (2017). An Investigation of the Relationship between Mathematics Performance of Students in a Non-Routine Problem, according to Grade and Gender. *International Journal of Innovation in Science and Mathematics Education*, 25(3).
- Hemmings, B., Grootenboer, P., & Kay, R. (2010). PREDICTING MATHEMATICS ACHIEVEMENT: THE INFLUENCE OF PRIOR ACHIEVEMENT AND ATTITUDES. *International Journal of Science and Mathematics Education*, 9(3), 691–705. <https://doi.org/10.1007/s10763-010-9224-5>
- Hennessy, S., & Dunham, P. (2002). EQUITY ISSUES AFFECTING MATHEMATICS LEARNING USING ICT. *Research in Mathematics Education*, 4(1), 145–165. <https://doi.org/10.1080/14794800008520107>
- Henra, K., Budayasa, I. K., & Ismail (2024). Revealing the dominant metacognitive activities of high school students in solving central tendency and dispersion problems based on gender. *Journal on Mathematics Education*, 15(4), 1357–1382. <https://doi.org/10.22342/jme.v15i4.pp1357-1382>
- Hernandez-Martinez, P., Dominguez, A., Zavala, G., Kambouri, M., Zubieta, J., & Clark, R. (2022). Applying the Delphi method with early-career researchers to explore a gender-issues agenda in STEM education. *Eurasia Journal of Mathematics, Science and Technology Education*, 18(11), em2177. <https://doi.org/10.29333/ejmste/12508>
- Hill, J. L., & Hunter, J. (2023). Examining the mathematics education values of diverse groups of students. *International Journal of Mathematical Education in Science and Technology*, 54(8), 1614–1633. <https://doi.org/10.1080/0020739x.2023.2184280>
- Hine, G., Forlin, C., & Chivers, P. (2023). Influences on student decisions to enrol in higher-level mathematics courses. *Mathematics Education Research Journal*, 36(3), 695–713. <https://doi.org/10.1007/s13394-023-00467-9>
- Hirst, K., & Meacock, S. (1999). Modular A-levels and Undergraduates. *Teaching Mathematics and its Applications: An International Journal of the IMA*, 18(3), 122–127. <https://doi.org/10.1093/teamat/18.3.122>

- Hite, R. L., & Milbourne, J. D. (2024). STEM teacher advocate identity in the United States. *School Science and Mathematics*, 125(5), 550–568. <https://doi.org/10.1111/ssm.12658>
- Ho, H. Z., Senturk, D., Lam, A. G., Zimmer, J. M., Hong, S., Okamoto, Y., Chiu, S. Y., Nakazawa, Y., & Wang, C. P. (2000). The Affective and Cognitive Dimensions of Math Anxiety: A Cross-National Study. *Journal for Research in Mathematics Education*, 31(3), 362–379. <https://doi.org/10.2307/749811>
- Holmes, K., Gore, J., Smith, M., & Lloyd, A. (2017). An Integrated Analysis of School Students' Aspirations for STEM Careers: Which Student and School Factors Are Most Predictive? *International Journal of Science and Mathematics Education*, 16(4), 655–675. <https://doi.org/10.1007/s10763-016-9793-z>
- Homer, M. S. (2018). Measuring mathematical affect in an international context: influences on pre-university attainment and degree choice. *Teaching Mathematics and its Applications: An International Journal of the IMA*, 38(1), 1–19. <https://doi.org/10.1093/teamat/hry003>
- HOMMIK, C., & LUIK, P. (2017). ADAPTING THE SURVEY OF ATTITUDES TOWARDS STATISTICS (SATS-36) FOR ESTONIAN SECONDARY SCHOOL STUDENTS. *STATISTICS EDUCATION RESEARCH JOURNAL*, 16(1), 228–239. <https://doi.org/10.52041/serj.v16i1.229>
- Hopkins, S., & Bayliss, D. (2017). The Prevalence and Disadvantage of Min-Counting in Seventh Grade: Problems with Confidence As Well As Accuracy? *Mathematical Thinking and Learning*, 19(1), 19–32. <https://doi.org/10.1080/10986065.2017.1258613>
- Hsu, S. K., & Hsu, Y. (2024). Supporting Young Learners in Learning Geometric Area Concepts Through Static Versus Dynamic Representation and Imagination Strategies. *International Journal of Science and Mathematics Education*, 23(2), 441–459. <https://doi.org/10.1007/s10763-024-10481-3>
- Huang, S. L., & Waxman, H. C. (1996). Classroom Observations of Middle School Students' Technology Use in Mathematics. *School Science and Mathematics*, 96(1), 28–34. <https://doi.org/10.1111/j.1949-8594.1996.tb10210.x>
- HUBER, S., REINHOLD, F., OBERSTEINER, A., & REISS, K. (2024). TEACHING STATISTICS WITH POSITIVE ORIENTATIONS BUT LITTLE KNOWLEDGE? TEACHERS' PROFESSIONAL COMPETENCE IN STATISTICS. *STATISTICS EDUCATION RESEARCH JOURNAL*, 23(1), 2. <https://doi.org/10.52041/serj.v23i1.610>
- Hunter, J., & Hill, J. L. (2024). Exploring mathematical wellbeing across cultures: insights from diverse students. *Mathematics Education Research Journal*, 37(3), 529–556. <https://doi.org/10.1007/s13394-024-00500-5>
- Hussein, Y. F., & Csíkos, C. (2023). The effect of teaching conceptual knowledge on students' achievement, anxiety about, and attitude toward mathematics. *Eurasia Journal of Mathematics, Science and Technology Education*, 19(2), em2226. <https://doi.org/10.29333/ejmste/12938>

- Huziak-Clark, T., Sondergeld, T., van Staaden, M., Knaggs, C., & Bullerjahn, A. (2015). Assessing the Impact of a Research-Based STEM Program on STEM Majors' Attitudes and Beliefs. *School Science and Mathematics*, 115(5), 226–236. <https://doi.org/10.1111/ssm.12118>
- Ifamuyiwa, S. A., & Akinsola, M. K. (2008). Improving senior secondary school students' attitude towards mathematics through self and cooperative-instructional strategies. *International Journal of Mathematical Education in Science and Technology*, 39(5), 569–585. <https://doi.org/10.1080/00207390801986874>
- Ikonen, K., Leinonen, R., Hirvonen, P. E., & Asikainen, M. A. (2019). Finnish Ninth Graders' Gender Appropriateness of Occupations. *EURASIA Journal of Mathematics, Science and Technology Education*, 15(12). <https://doi.org/10.29333/ejmste/111995>
- Ing, M. (2013). GENDER DIFFERENCES IN THE INFLUENCE OF EARLY PERCEIVED PARENTAL SUPPORT ON STUDENT MATHEMATICS AND SCIENCE ACHIEVEMENT AND STEM CAREER ATTAINMENT. *International Journal of Science and Mathematics Education*, 12(5), 1221–1239. <https://doi.org/10.1007/s10763-013-9447-3>
- Innabi, H., & Dodeen, H. (2006). Content Analysis of Gender-related Differential Item Functioning TIMSS Items in Mathematics in Jordan. *School Science and Mathematics*, 106(8), 328–337. <https://doi.org/10.1111/j.1949-8594.2006.tb17753.x>
- Innabi, H., & Dodeen, H. (2018). Gender differences in mathematics achievement in Jordan: A differential item functioning analysis of the 2015 TIMSS. *School Science and Mathematics*, 118(3-4), 127–137. <https://doi.org/10.1111/ssm.12269>
- Isiksal, M., & Cakiroghi, E. (2008). Gender Differences Regarding Mathematics Achievement: The Case of Turkish Middle School Students. *School Science and Mathematics*, 108(3), 113–120. <https://doi.org/10.1111/j.1949-8594.2008.tb17814.x>
- Ismail, N. A., & Awang, H. (2007). Differentials in Mathematics Achievement among Eighth-Grade Students in Malaysia. *International Journal of Science and Mathematics Education*, 6(3), 559–571. <https://doi.org/10.1007/s10763-007-9109-4>
- Izagirre, A., Anasagasti, J., & Berciano, A. (2025). Determinants of secondary school teachers' attitudes towards probability and its teaching. *STATISTICS EDUCATION RESEARCH JOURNAL*, 24(2), 2. <https://doi.org/10.52041/serj.v24i2.751>
- Jackson, C., Buchheister, K., & Taylor, C. E. (2023). Attending to what prospective teachers notice about students' intersecting identities. *School Science and Mathematics*, 123(8), 461–475. <https://doi.org/10.1111/ssm.12609>
- Jackson, C., Wilhelm, J. A., Lamar, M., & Cole, M. (2015). Gender and Racial Differences: Development of Sixth Grade Students' Geometric Spatial Visualization within an Earth/Space Unit. *School Science and Mathematics*, 115(7), 330–343. <https://doi.org/10.1111/ssm.12136>

- Jacobs, G. J., & Durandt, R. (2016). Attitudes of Pre-Service Mathematics Teachers towards Modelling: A South African Inquiry. *EURASIA Journal of Mathematics, Science and Technology Education*, 13(1). <https://doi.org/10.12973/eurasia.2017.00604a>
- Jamil, F. M., Larsen, R. A., & Hamre, B. K. (2018). Exploring Longitudinal Changes in Teacher Expectancy Effects on Children's Mathematics Achievement. *Journal for Research in Mathematics Education*, 49(1), 57–90. <https://doi.org/10.5951/jresmetheduc.49.1.0057>
- Jao, L., Hall, J., & Di Placido, C. (2023). "If you're a dude, you're a chick, whatever the hell in between, you need to know about maths": the Australian and Canadian general public's views of gender and mathematics. *Mathematics Education Research Journal*, 36(2), 339–365. <https://doi.org/10.1007/s13394-023-00448-y>
- Jaremus, F. (2020). When girls do masculinity like boys do: establishing gender heteroglossia in school mathematics participation. *Mathematics Education Research Journal*, 33(4), 713–731. <https://doi.org/10.1007/s13394-020-00355-6>
- Jaremus, F., Gore, J., Fray, L., & Prieto-Rodriguez, E. (2018). Senior secondary student participation in STEM: Beyond national statistics. *Mathematics Education Research Journal*, 31(2), 151–173. <https://doi.org/10.1007/s13394-018-0247-5>
- Jeffries, D., Curtis, D. D., & Conner, L. N. (2019). Student Factors Influencing STEM Subject Choice in Year 12: a Structural Equation Model Using PISA/LSAY Data. *International Journal of Science and Mathematics Education*, 18(3), 441–461. <https://doi.org/10.1007/s10763-019-09972-5>
- Jenßen, L., Roesken-Winter, B., & Blömeke, S. (2022). Measuring Pre-service Primary Teachers' Shame in Mathematics—a Comprehensive Validation Study. *International Journal of Science and Mathematics Education*, 21(2), 463–488. <https://doi.org/10.1007/s10763-022-10253-x>
- Johns, C. (2020). Self-Regulation in First-Semester Calculus. *International Journal of Research in Undergraduate Mathematics Education*, 6(3), 404–420. <https://doi.org/10.1007/s40753-020-00114-9>
- Johnson, E., Andrews-Larson, C., Keene, K., Melhuish, K., Keller, R., & Fortune, N. (2020). Inquiry and Gender Inequity in the Undergraduate Mathematics Classroom. *Journal for Research in Mathematics Education*, 51(4), 504–516. <https://doi.org/10.5951/jresmetheduc-2020-0043>
- Johnson, M., & Kuennen, E. (2006). Basic Math Skills and Performance in an Introductory Statistics Course. *Journal of Statistics Education*, 14(2). <https://doi.org/10.1080/10691898.2006.11910581>
- Jones, L., & Magill, K. (2023). A Critical Examination of the Relationship Between Social and Personal Constructions of Gender and Race and their Impact on Mathematical Identity. *Investigations in Mathematics Learning*, 15(4), 348–366. <https://doi.org/10.1080/19477503.2023.2254152>

- Joy, A., Mathews, C. J., Hartstone-Rose, A., & Mulvey, K. L. (2024). What does a scientist look like? Children's perceptions of scientist gender and skin tone. *School Science and Mathematics*, 125(6), 631–643. <https://doi.org/10.1111/ssm.18308>
- Julie, C. (2020). Modelling competencies of school learners in the beginning and final year of secondary school mathematics. *International Journal of Mathematical Education in Science and Technology*, 51(8), 1181–1195. <https://doi.org/10.1080/0020739x.2020.1725165>
- Jungwirth, H. (2008). On the role of computers and complementary situations for gendering in mathematics classrooms. *ZDM*, 40(4), 579–590. <https://doi.org/10.1007/s11858-008-0116-x>
- Jurdak, M., & Abu Zein, R. (1998). The Effect of Journal Writing on Achievement in and Attitudes Toward Mathematics. *School Science and Mathematics*, 98(8), 412–419. <https://doi.org/10.1111/j.1949-8594.1998.tb17433.x>
- Kaino, L. M. (1998). Some Swaziland High School Students' Attitudes towards the Study of Mathematics. *African Journal of Research in Mathematics Science and Technology Education*, 33–40. <https://doi.org/10.1080/10288457.1998.10756097>
- Kaiser, G., & Steisel, T. (2000). Results of an analysis of the TIMS study from a gender perspective. *Zentralblatt für Didaktik der Mathematik*, 32(1), 18–24. <https://doi.org/10.1007/bf02652735>
- Kaiser, G., & Zhu, Y. (2022). Gender differences in mathematics achievement: A secondary analysis of Programme for International Student Assessment data from Shanghai. *Asian Journal for Mathematics Education*, 1(1), 115–130. <https://doi.org/10.1177/27527263221091373>
- Kaleli-Yilmaz, G., & Hanci, A. (2015). Examination of the 8th grade students' TIMSS mathematics success in terms of different variables1. *International Journal of Mathematical Education in Science and Technology*, 47(5), 674–695. <https://doi.org/10.1080/0020739x.2015.1102977>
- Kanyongo, G. Y., Schreiber, J. B., & Brown, L. I. (2007). Factors affecting mathematics achievement among 6th graders in three sub-Saharan African countries: The use of hierarchical linear models (HLM). *African Journal of Research in Mathematics, Science and Technology Education*, 11(1), 37–46. <https://doi.org/10.1080/10288457.2007.10740610>
- KAPITANOFF, S., & PANDEY, C. (2018). COLLABORATIVE TESTING IN STATISTICS: GROUP INTERACTION, ANXIETY, AND CLASS PERFORMANCE. *STATISTICS EDUCATION RESEARCH JOURNAL*, 17(2), 51–67. <https://doi.org/10.52041/serj.v17i2.158>
- Karjanto, N. (2017). Attitude toward mathematics among the students at Nazarbayev University Foundation Year Programme. *International Journal of Mathematical Education in Science and Technology*, 48(6), 849–863. <https://doi.org/10.1080/0020739x.2017.1285060>

- Kayan Fadlelmula, F. (2013). Pre-service Mathematics Teachers' Anxieties toward Teaching Mathematics in Real Classroom Settings. *The International Journal of Science, Mathematics and Technology Learning*, 19(1), 37–47. <https://doi.org/10.18848/2327-7971/cgp/v19i01/49048>
- Kayan Fadlelmula, F., Sellami, A., & Le, K. (2022). STEM learning during the COVID-19 pandemic in Qatar: Secondary school students' and teachers' perspectives. *Eurasia Journal of Mathematics, Science and Technology Education*, 18(6), em2123. <https://doi.org/10.29333/ejmste/12102>
- Kazima, M. (2013). Students' Preferences for Contexts and their Relevance to School Mathematics in Malawi. *African Journal of Research in Mathematics, Science and Technology Education*, 17(3), 244–254. <https://doi.org/10.1080/10288457.2013.848537>
- Kelly, B., Devlin, M., Giffin, T., & Smith, L. (2022). Family Learning Online During Lockdown in the UK. *Adults Learning Mathematics International Journal*, 16(1 Special issue), 20–35.
- Kërënxhi, S., & Gjoci, P. (2014). DUAL TREATMENTS AS STARTING POINT FOR INTEGRATIVE PERCEPTIONS IN TEACHING MATHEMATICS. *International Journal of Science and Mathematics Education*, 13(4), 793–809. <https://doi.org/10.1007/s10763-014-9581-6>
- Kersey, E., & Voigt, M. (2020). Finding community and overcoming barriers: experiences of queer and transgender postsecondary students in mathematics and other STEM fields. *Mathematics Education Research Journal*, 33(4), 733–756. <https://doi.org/10.1007/s13394-020-00356-5>
- Kim, H., & Jeong, S. (. K. (2015). The Effect of a Climate Change Monitoring Program on Students' Knowledge and Perceptions of STEAM Education in the Republic of Korea. *EURASIA Journal of Mathematics, Science and Technology Education*, 11(6). <https://doi.org/10.12973/eurasia.2015.1390a>
- Kirkham, J., Chapman, E., & Wildy, H. (2019). Factors considered by Western Australian Year 10 students in choosing Year 11 mathematics courses. *Mathematics Education Research Journal*, 32(4), 719–741. <https://doi.org/10.1007/s13394-019-00277-y>
- Kismiantini, K., Setiawan, E. P., Pierewan, A. C., & Montesinos-Lopez, O. A. (2021). GROWTH MINDSET, SCHOOL CONTEXT, AND MATHEMATICS ACHIEVEMENT IN INDONESIA: A MULTILEVEL MODEL. *Journal on Mathematics Education*, 12(2), 279–294. <https://doi.org/10.22342/jme.12.2.13690.279-294>
- Kitchen, R. (2022). The power of their ideas: leveraging teachers' mathematical ideas in professional development. *International Journal of Mathematical Education in Science and Technology*, 53(7), 1835–1858. <https://doi.org/10.1080/0020739x.2020.1847337>
- Kiwanuka, H. N., Van Damme, J., Van den Noortgate, W., & Reynolds, C. (2022). Temporal relationship between attitude toward mathematics and mathematics achievement. *International Journal of Mathematical Education in Science and Technology*, 53(6), 1546–1570. <https://doi.org/10.1080/0020739x.2020.1832268>

- Klemer, A., & Rapoport, S. (2020). Origami and GeoGebra Activities Contribute to Geometric Thinking in Second Graders. *Eurasia Journal of Mathematics, Science and Technology Education*, 16(11), em1894. <https://doi.org/10.29333/ejmste/8537>
- Kloosterman, P., Tassell, J. (., Ponniah, A. G., & Essex, N. K. (2008). Perceptions of Mathematics and Gender. *School Science and Mathematics*, 108(4), 149–162. <https://doi.org/10.1111/j.1949-8594.2008.tb17821.x>
- Knabbe, A., Leiss, D., & Ehmke, T. (2024). Reality-Based Tasks with Complex-Situations: Identifying Sociodemographic and Cognitive Factors for Solution. *International Journal of Science and Mathematics Education*, 23(1), 97–120. <https://doi.org/10.1007/s10763-024-10463-5>
- Knowles, J. M. (2008). Gender in mathematics relationality: counseling underprepared college students. *ZDM*, 40(4), 673–692. <https://doi.org/10.1007/s11858-008-0099-7>
- Kokka, K. (2024). “I really got to think about my background, their background, and how do we come together on something?”: One emergent mathematics teacher leader’s reflexive journey with Social Justice Mathematics. *School Science and Mathematics*, 125(5), 446–461. <https://doi.org/10.1111/ssm.12688>
- Koller, O., Baumert, J., & Schnabel, K. (2001). Does Interest Matter? The Relationship between Academic Interest and Achievement in Mathematics. *Journal for Research in Mathematics Education*, 32(5), 448. <https://doi.org/10.2307/749801>
- Kolovou, A., van den Heuvel-Panhuizen, M., & Köller, O. (2013). An Intervention Including an Online Game to Improve Grade 6 Students’ Performance in Early Algebra. *Journal for Research in Mathematics Education*, 44(3), 510–549. <https://doi.org/10.5951/jresmetheduc.44.3.0510>
- Kolovou, M., Ran, H., & Secada, W. (2022). CGI Teachers’ Patterns of Interacting with Male and Female Students During Their First and Second Years of Practice. *International Journal of Science and Mathematics Education*, 21(5), 1451–1472. <https://doi.org/10.1007/s10763-022-10314-1>
- Kombe, D., Che, S. M., & Bridges, W. (2019). Students’ gendered perceptions of mathematics in middle grades single-sex and coeducational classrooms. *School Science and Mathematics*, 119(7), 417–427. <https://doi.org/10.1111/ssm.12363>
- Kombe, D., Che, S. M., Carter, T. L., & Bridges, W. (2016). Student Academic Self-Concept and Perception of Classroom Environment in Single-Sex and Coeducational Middle Grades Mathematics Classes. *School Science and Mathematics*, 116(5), 265–275. <https://doi.org/10.1111/ssm.12178>
- Korpershoek, H., Kuyper, H., & van der Werf, G. (2014). THE RELATION BETWEEN STUDENTS’ MATH AND READING ABILITY AND THEIR MATHEMATICS, PHYSICS, AND CHEMISTRY EXAMINATION GRADES IN SECONDARY EDUCATION. *International Journal of Science and Mathematics Education*, 13(5), 1013–1037. <https://doi.org/10.1007/s10763-014-9534-0>

- Kotsopoulos, D., Weatherby, C., & Woolford, D. G. (2021). Using guided notes to support learning in first-year calculus. *International Journal of Mathematical Education in Science and Technology*, 53(6), 1629–1644. <https://doi.org/10.1080/0020739x.2021.1910742>
- Kovács, T., & Hanjra, G. (2025). Current obstacles in global STEM education and a possible practical solution to overcome these. *Eurasia Journal of Mathematics, Science and Technology Education*, 21(12), em2747. <https://doi.org/10.29333/ejmste/17493>
- Kung, H. Y., & Lee, C. Y. (2018). Math Self-Concept and Mathematics Achievement: Examining Gender Variation and Reciprocal Relations among Junior High School Students in Taiwan. *EURASIA Journal of Mathematics, Science and Technology Education*, 14(4). <https://doi.org/10.29333/ejmste/82535>
- Kunwar, R., & Sharma, L. (2020). Exploring Teachers' Knowledge and Students' Status about Dyscalculia at Basic Level Students in Nepal. *Eurasia Journal of Mathematics, Science and Technology Education*, 16(12), em1906. <https://doi.org/10.29333/ejmste/8940>
- Kunwar, R., G. C., L., Dahal, P. P., & Thapa, K. B. (2024). Unveiling the silent struggle: Exploring the influence of dyslexia on elementary students' mathematics learning in Nepal. *Eurasia Journal of Mathematics, Science and Technology Education*, 20(7), em2474. <https://doi.org/10.29333/ejmste/14752>
- Kurnia, A. B., Lowrie, T., & Patahuddin, S. M. (2023). The development of high school students' statistical literacy across grade level. *Mathematics Education Research Journal*, 36(S1), 7–35. <https://doi.org/10.1007/s13394-023-00449-x>
- Kyaw, K. M., & Vidákovich, T. (2025). The relationship between spatial reasoning and geometric reasoning in teachers. *Eurasia Journal of Mathematics, Science and Technology Education*, 21(8), em2684. <https://doi.org/10.29333/ejmste/16718>
- Kyeremeh, P., Kwadwo Awuah, F., & Clark Orey, D. (2025). Challenges regarding the integration of ethnomathematical perspectives into geometry teaching: The faculty reflection. *Asian Journal for Mathematics Education*, 4(3), 416–430. <https://doi.org/10.1177/27527263251326330>
- Landa, J., Berciano, A., & Marbán, J. M. (2025). Self-regulation of primary school teachers in initial training when solving mathematical problems in cooperative learning contexts. *International Electronic Journal of Mathematics Education*, 20(2), em0813. <https://doi.org/10.29333/iejme/15817>
- Landim, C., & FitzSimons, G. E. (2022). The Brazilian Public Schools Math Olympics (OBMEP): 15 years promoting social mobility through academic achievement. *ZDM – Mathematics Education*, 54(5), 1101–1113. <https://doi.org/10.1007/s11858-022-01341-x>
- Lane, C., Styne, M., & O'Donoghue, J. (2016). Post-primary students' images of mathematics: findings from a survey of Irish ordinary level mathematics students. *International Journal of Mathematical Education in Science and Technology*, 47(7), 1009–1027. <https://doi.org/10.1080/0020739x.2016.1170899>

- Langer-Osuna, J. M. (2011). How Brianna Became Bossy and Kofi Came Out Smart: Understanding the Trajectories of Identity and Engagement for Two Group Leaders in a Project-Based Mathematics Classroom. *Canadian Journal of Science, Mathematics and Technology Education*, 11(3), 207–225. <https://doi.org/10.1080/14926156.2011.595881>
- Lasater, R. S., Joseph, A., & Cummiskey, K. (2022). Exploring a complex gender wage gap dataset: an introductory activity in identifying issues and data visualization. *Teaching Statistics*, 45(1), 14–21. <https://doi.org/10.1111/test.12319>
- Laursen, S. L., Hassi, M. L., Kogan, M., & Weston, T. J. (2014). Benefits for Women and Men of Inquiry-Based Learning in College Mathematics: A Multi-Institution Study. *Journal for Research in Mathematics Education*, 45(4), 406–418. <https://doi.org/10.5951/jresmetheduc.45.4.0406>
- Lawrenz, F., Gravely, A., & Ooms, A. (2006). Perceived Helpfulness and Amount of Use of Technology in Science and Mathematics Classes at Different Grade Levels. *School Science and Mathematics*, 106(3), 133–139. <https://doi.org/10.1111/j.1949-8594.2006.tb18170.x>
- Lazarides, R., & Rubach, C. (2017). Instructional characteristics in mathematics classrooms: relationships to achievement goal orientation and student engagement. *Mathematics Education Research Journal*, 29(2), 201–217. <https://doi.org/10.1007/s13394-017-0196-4>
- Leder, G. (2011). Mathematics Taught Me Einstein’s Old Cocktail of Inspiration and Perspiration: Mathematically Talented Teenagers as Adults. *Canadian Journal of Science, Mathematics and Technology Education*, 11(1), 29–38. <https://doi.org/10.1080/14926156.2011.548897>
- Leder, G. C., & Forgasz, H. J. (1997). Single-Sex Classes in a Coeducational High School Highlighting Parents’ Perspectives. *Mathematics Education Research Journal*, 9(3), 274–291. <https://doi.org/10.1007/bf03217319>
- Leder, G. C., & Forgasz, H. J. (2018). Measuring who counts: gender and mathematics assessment. *ZDM*, 50(4), 687–697. <https://doi.org/10.1007/s11858-018-0939-z>
- Leder, G. C., & Taylor, P. (1995). Achievement in the Australian Mathematics Competition: A question of age? *Mathematics Education Research Journal*, 7(2), 95–110. <https://doi.org/10.1007/bf03217279>
- Lee, C. Y., & Yuan, Y. (2010). GENDER DIFFERENCES IN THE RELATIONSHIP BETWEEN TAIWANESE ADOLESCENTS’ MATHEMATICS ATTITUDES AND THEIR PERCEPTIONS TOWARD VIRTUAL MANIPULATIVES. *International Journal of Science and Mathematics Education*, 8(5), 937–950. <https://doi.org/10.1007/s10763-010-9193-8>
- Lee, Y., Capraro, R. M., & Bicer, A. (2019). Gender difference on spatial visualization by college students’ major types as STEM and non-STEM: a meta-analysis. *International Journal of Mathematical Education in Science and Technology*, 50(8), 1241–1255. <https://doi.org/10.1080/0020739x.2019.1640398>

- Leedy, M. G., LaLonde, D., & Runk, K. (2003). Gender Equity in Mathematics: Beliefs of Students, Parents, and Teachers. *School Science and Mathematics*, 103(6), 285–292. <https://doi.org/10.1111/j.1949-8594.2003.tb18151.x>
- Lembke, E., Lee, Y. S., Park, Y. S., & Hampton, D. (2016). Longitudinal growth on curriculum-based measurements mathematics measures for early elementary students. *ZDM*, 48(7), 1049–1063. <https://doi.org/10.1007/s11858-016-0804-x>
- Leonard, J., Walker, E. N., Bloom, V. R., & Joseph, N. M. (2020). Mathematics Literacy, Identity Resilience, and Opportunity Sixty Years Since Brown v. Board. *Journal of Urban Mathematics Education*, 13(1B). <https://doi.org/10.21423/jume-v13i1ba405>
- Lesser, L. M., Wall, A. A., Carver, R. H., Pearl, D. K., Martin, N., Kuiper, S., Posner, M. A., Erickson, P., Liao, S. M., Albert, J., & Weber, J. J. (2013). Using Fun in the Statistics Classroom: An Exploratory Study of College Instructors' Hesitations and Motivations. *Journal of Statistics Education*, 21(1). <https://doi.org/10.1080/10691898.2013.11889659>
- Leyva, L. A. (2021). Black Women's Counter-Stories of Resilience and Within-Group Tensions in the White, Patriarchal Space of Mathematics Education. *Journal for Research in Mathematics Education*, 52(2), 117–151. <https://doi.org/10.5951/jresmetheduc-2020-0027>
- Leyva, L. A., Amman, K., Wolf McMichael, E. A., Igbinosun, J., & Khan, N. (2022). Support for All? Confronting Racism and Patriarchy to Promote Equitable Learning Opportunities through Undergraduate Calculus Instruction. *International Journal of Research in Undergraduate Mathematics Education*, 8(2), 339–364. <https://doi.org/10.1007/s40753-022-00177-w>
- Li, C., & Nuttall, R. (2001). Writing Chinese and mathematics achievement: A study with Chinese-American undergraduates. *Mathematics Education Research Journal*, 13(1), 15–27. <https://doi.org/10.1007/bf03217096>
- Li, M., Vale, C., Tan, H., & Blannin, J. (2025). Exploring demographic influences on digital technology integration in Chinese primary mathematics education. *International Electronic Journal of Mathematics Education*, 20(2), em0810. <https://doi.org/10.29333/iejme/15814>
- Lim, J. H. (2008). Adolescent girls' construction of moral discourses and appropriation of primary identity in a mathematics classroom. *ZDM*, 40(4), 617–631. <https://doi.org/10.1007/s11858-008-0119-7>
- Lim, J. H., Chae, J., Schinck-Mikel, A. G., & Watson, J. (2013). Varied Meanings and Engagement in School Mathematics: Cross-Case Analysis of Three High-Achieving Young Adolescent Girls. *School Science and Mathematics*, 113(4), 191–200. <https://doi.org/10.1111/ssm.12017>
- Lin, K. Y., & Williams, P. J. (2015). Taiwanese Preservice Teachers' Science, Technology, Engineering, and Mathematics Teaching Intention. *International Journal of Science and Mathematics Education*, 14(6), 1021–1036. <https://doi.org/10.1007/s10763-015-9645-2>

- Lin, S. H., Ma, H. L., Lee, D. C., & Wu, D. B. (2015). A Study of Van Hiele of Geometric Thinking among 1st through 6th Graders. *EURASIA Journal of Mathematics, Science and Technology Education*, 11(5). <https://doi.org/10.12973/eurasia.2015.1412a>
- Lippiello, S., Carli, M., & Pantano, O. (2025). The Relationship between Performance and Confidence in the Secondary-School Version of the Test of Calculus and Vectors in Mathematics and Physics. *International Journal of Science and Mathematics Education*, 23(7), 2997–3020. <https://doi.org/10.1007/s10763-025-10560-z>
- Litster, K., Lommatsch, C. W., Novak, J. R., Moyer-Packenham, P. S., Harmon, M. J., Roxburgh, A. L., & Bullock, E. P. (2020). The Role of Gender on the Associations among Children's Attitudes, Mathematics Knowledge, Digital Game Use, Perceptions of Affordances, and Achievement. *International Journal of Science and Mathematics Education*, 19(7), 1463–1483. <https://doi.org/10.1007/s10763-020-10111-8>
- Liu, X., Wachira, P., Koc, S., & Pourdavood, R. (2021). An Exploratory Study of Predictors of Pre-Service Teachers' Intention to Integrate Computer Games in Mathematics Education. *International Journal of Education in Mathematics, Science and Technology*, 10(1), 145–161. <https://doi.org/10.46328/ijemst.1827>
- Llewellyn, A. (2009). 'Gender games': a post-structural exploration of the prospective teacher, mathematics and identity. *Journal of Mathematics Teacher Education*, 12(6), 411–426. <https://doi.org/10.1007/s10857-009-9109-0>
- Logan, T., & Lowrie, T. (2017). Gender perspectives on spatial tasks in a national assessment: a secondary data analysis. *Research in Mathematics Education*, 19(2), 199–216. <https://doi.org/10.1080/14794802.2017.1334577>
- Lorah, J. A., & Valdivia, M. (2020). Diversity in Statistics Education at Postsecondary Institutions. *International Journal of Research in Undergraduate Mathematics Education*, 7(1), 21–32. <https://doi.org/10.1007/s40753-020-00120-x>
- Lorah, J., & Miller, A. (2023). Educational Aspirations of Diverse Groups among Undergraduate Statistics Majors. *Journal of Urban Mathematics Education*, 16(2), 14–30. <https://doi.org/10.21423/jume-v16i2a548>
- Louis, R. A., & Mistele, J. M. (2011). THE DIFFERENCES IN SCORES AND SELF-EFFICACY BY STUDENT GENDER IN MATHEMATICS AND SCIENCE. *International Journal of Science and Mathematics Education*, 10(5), 1163–1190. <https://doi.org/10.1007/s10763-011-9325-9>
- Lowrie, T., & Jorgensen, R. (2017). Equity and spatial reasoning: reducing the mathematical achievement gap in gender and social disadvantage. *Mathematics Education Research Journal*, 30(1), 65–75. <https://doi.org/10.1007/s13394-017-0213-7>
- Lubienski, S. T., Ganley, C. M., Makowski, M. B., Miller, E. K., & Timmer, J. D. (2021). "Bold Problem Solving": A New Construct for Understanding Gender Differences in Mathematics. *Journal for Research in Mathematics Education*, 52(1), 12–61. <https://doi.org/10.5951/jresmetheduc-2020-0136>

- Lubienski, S. T., Robinson, J. P., Crane, C. C., & Ganley, C. M. (2013). BRIEF REPORT: Girls' and Boys' Mathematics Achievement, Affect, and Experiences: Findings From ECLS-K. *Journal for Research in Mathematics Education*, 44(4), 634–645. <https://doi.org/10.5951/jresmetheduc.44.4.0634>
- Lubienski, S., & Ataide Pinheiro, W. (2020). Gender and Mathematics: What Can Other Disciplines Tell Us? What is Our Role? *Journal of Urban Mathematics Education*, 13(1). <https://doi.org/10.21423/jume-v13i1a377>
- Lutes, A. C., Hsiao, E. L., Downey, S., Kim, D., & Hartsell, T. S. (2021). The Effect of a Hip-Hop Based Education Unit on Fifth-Grade Students' Math Achievements. *The International Journal of Science, Mathematics and Technology Learning*, 28(1), 61–77. <https://doi.org/10.18848/2327-7971/cgp/v28i01/61-77>
- Lutfiyya, L. A. (1998). Mathematical thinking of high school students in Nebraska. *International Journal of Mathematical Education in Science and Technology*, 29(1), 55–64. <https://doi.org/10.1080/0020739980290106>
- Lyakhova, S., & Joubert, M. (2021). Post-16 Further Mathematics blended learning: learner self-regulation, mathematical resilience and technology. *Teaching Mathematics and its Applications: An International Journal of the IMA*, 41(1), 51–68. <https://doi.org/10.1093/teamat/hrab005>
- Ma, X. (1999). A Meta-Analysis of the Relationship between Anxiety toward Mathematics and Achievement in Mathematics. *Journal for Research in Mathematics Education*, 30(5), 520. <https://doi.org/10.2307/749772>
- Ma, X., & Kishor, N. (1997). Assessing the Relationship between Attitude toward Mathematics and Achievement in Mathematics: A Meta-Analysis. *Journal for Research in Mathematics Education*, 28(1), 26. <https://doi.org/10.2307/749662>
- MacArthur, K. R., & Santo, J. B. (2022). A Multi-Level Analysis of the Effects of Statistics Anxiety/Attitudes on Trajectories of Exam Scores. *Journal of Statistics and Data Science Education*, 31(1), 102–112. <https://doi.org/10.1080/26939169.2022.2093805>
- Mahaur, N., Ahuja, S., & Srivastava, A. (2024). Analysis of Students' Errors in Solving Mathematics Problems on the Semester Final Examination. *The International Journal of Science, Mathematics and Technology Learning*, 32(1), 173–191. <https://doi.org/10.18848/2327-7971/cgp/v32i01/173-191>
- Manalu, M. S., & Chang, C. Y. (2025). Unlocking Indonesian primary students' attitudes toward STEM education and interests in STEM-related careers using latent profile analysis. *Eurasia Journal of Mathematics, Science and Technology Education*, 21(2), em2584. <https://doi.org/10.29333/ejmste/15954>
- Mandrikas, A., & Stefanidou, C. (2025). Greek primary teachers' views about STEM education. *European Journal of Science and Mathematics Education*, 13(2), 119–136. <https://doi.org/10.30935/scimath/16162>

- Mansour, N., Said, Z., & Abu-Tineh, A. (2024). Factors impacting science and mathematics teachers' competencies and self-efficacy in TPACK for PBL and STEM. *Eurasia Journal of Mathematics, Science and Technology Education*, 20(5), em2442. <https://doi.org/10.29333/ejmste/14467>
- Maphutha, K., Maoto, S., & Kibirige, I. (2022). The effect of the activity-based approach on grade 11 learners' performance in solving two-dimensional trigonometric problems. *Eurasia Journal of Mathematics, Science and Technology Education*, 18(10), em2155. <https://doi.org/10.29333/ejmste/12405>
- Marange, I. Y., & Tatira, B. (2024). Gender dynamics in GeoGebra integration: In-service mathematics teachers' development. *Eurasia Journal of Mathematics, Science and Technology Education*, 20(6), em2457. <https://doi.org/10.29333/ejmste/14629>
- Marbán, J. M., & Mulenga, E. M. (2022). Social Media Usage Among University Students of Mathematics Education in Zambia: A Quantitative Analysis Approach. *International Journal for Technology in Mathematics Education*, 29(4), 201–215. https://doi.org/10.1564/tme_v29.4.02
- Marcone, R., Bose, A., Milani, R., & Kumar, V. (2024). Non-Typical learning sites: A railway platform and an interplay between foreground, uncertainty, and hope. *Bolema: Boletim de Educação Matemática*, 38. <https://doi.org/10.1590/1980-4415v38a240065>
- Maričić, S. M., & Stamatović, J. D. (2017). The Effect of Preschool Mathematics Education in Development of Geometry Concepts in Children. *EURASIA Journal of Mathematics, Science and Technology Education*, 13(9). <https://doi.org/10.12973/eurasia.2017.01057a>
- Marin Ballon, E. M., Romero Gomez, F. L., Linares Flores Castro, A. E., & Flores Castro Linares, M. R. (2024). Evaluating problem-solving and procedural skills of first-year students in a Peruvian higher education institution. *Eurasia Journal of Mathematics, Science and Technology Education*, 20(2), em2396. <https://doi.org/10.29333/ejmste/14154>
- MARTIN, N., HUGHES, J., & FUGELSANG, J. (2017). THE ROLES OF EXPERIENCE, GENDER, AND INDIVIDUAL DIFFERENCES IN STATISTICAL REASONING. *STATISTICS EDUCATION RESEARCH JOURNAL*, 16(2), 454–475. <https://doi.org/10.52041/serj.v16i2.201>
- Marzocchi, A. S., Martinez, P. M., & Truong, A. P. (2023). An examination of the human presence in precalculus textbooks: What are we communicating to students about who uses mathematics and how? *School Science and Mathematics*, 123(4-5), 184–200. <https://doi.org/10.1111/ssm.12592>
- Mazza, L., & Gambini, A. (2022). The phenomenon of the gender gap among gifted students: the situation in Italy based on analysis of results in mathematics competitions. *Teaching Mathematics and its Applications: An International Journal of the IMA*, 42(2), 170–186. <https://doi.org/10.1093/teamat/hrac009>
- McCullagh, O., Ryan, M., & Fitzmaurice, O. (2024). Mathematics anxiety in undergraduate business studies students. *Teaching Mathematics and its Applications: An International Journal of the IMA*, 43(2), 125–146. <https://doi.org/10.1093/teamat/hrae001>

- McCulloch, A. W., Kenney, R. H., & Keene, K. A. (2013). What to Trust: Reconciling Mathematical Work Done by Hand with Conflicting Graphing Calculator Solutions. *School Science and Mathematics*, 113(4), 201–210. <https://doi.org/10.1111/ssm.12016>
- McINTEE, S. E., GOULET-PELLETIER, J. C., WILLIOT, A., DECK-LÉGER, E., LALANDE, D., CANTINOTTI, M., & COUSINEAU, D. (2022). (MAL)ADAPTIVE COGNITIONS AS PREDICTORS OF STATISTICS ANXIETY. *STATISTICS EDUCATION RESEARCH JOURNAL*, 21(1), 5. <https://doi.org/10.52041/serj.v21i1.364>
- Mejía-Rodríguez, A. M., Luyten, H., & Meelissen, M. R. M. (2020). Gender Differences in Mathematics Self-concept Across the World: an Exploration of Student and Parent Data of TIMSS 2015. *International Journal of Science and Mathematics Education*, 19(6), 1229–1250. <https://doi.org/10.1007/s10763-020-10100-x>
- Meng, C. C., Idris, N., & Eu, L. K. (2014). Secondary Students' Perceptions of Assessments in Science, Technology, Engineering, and Mathematics (STEM). *EURASIA Journal of Mathematics, Science and Technology Education*, 10(3). <https://doi.org/10.12973/eurasia.2014.1070a>
- Miao, Z., Bokhove, C., Reynolds, D., & Charalambous, C. Y. (2022). Rational numbers and proportional reasoning in Chinese primary schools: Patterns, latent classes, and reasoning processes. *Asian Journal for Mathematics Education*, 1(4), 408–436. <https://doi.org/10.1177/27527263221143802>
- Milovanović, I., & Branovački, B. (2020). Adaptation and Psychometric Evaluation of Modified Abbreviated Math Anxiety Scale for Children in Serbia. *International Journal of Science and Mathematics Education*, 19(3), 579–598. <https://doi.org/10.1007/s10763-020-10066-w>
- Mittelberg, D., & Ari, L. L. (1997). Gender differences in mathematics among jewish and Arab youth in Israel. *Mathematics Education Research Journal*, 9(3), 347–351. <https://doi.org/10.1007/bf03217324>
- Moffatt, L., Anderson, A., Anderson, J., & Shapiro, J. (2009). Gender and Mathematics at Play: Parents' Constructions of their Preschoolers Mathematical Capabilities. *Investigations in Mathematics Learning*, 2(1), 1–25. <https://doi.org/10.1080/24727466.2009.11790288>
- Mohamad Hasim, S., Rosli, R., & Halim, L. (2024). A quantitative case study of secondary school students' level of statistical thinking. *Eurasia Journal of Mathematics, Science and Technology Education*, 20(4), em2421. <https://doi.org/10.29333/ejmste/14358>
- Mohamed, R. H., Khalil, I. A., & Awaji, B. M. (2023). Mathematics teachers' awareness of effective teaching practices: A comparative study. *Eurasia Journal of Mathematics, Science and Technology Education*, 19(2), em2230. <https://doi.org/10.29333/ejmste/12962>
- Morrow, C., & Schowengerdt, I. (2008). Stepping beyond high school mathematics: a case study of high school women. *ZDM*, 40(4), 693–708. <https://doi.org/10.1007/s11858-008-0108-x>
- Mujtaba, T., & Reiss, M. J. (2015). The Millennium Development Goals Agenda: Constraints of Culture, Economy, and Empowerment in Influencing the Social Mobility of Pakistani Girls on Mathematics and Science Related Higher Education Courses in Universities in Pakistan.

Canadian Journal of Science, Mathematics and Technology Education, 15(1), 51–68.
<https://doi.org/10.1080/14926156.2014.992556>

- Mujtaba, T., & Reiss, M. J. (2016). “I Fall Asleep in Class ... But Physics Is Fascinating”: The Use of Large-Scale Longitudinal Data to Explore the Educational Experiences of Aspiring Girls in Mathematics and Physics. *Canadian Journal of Science, Mathematics and Technology Education*, 16(4), 313–330. <https://doi.org/10.1080/14926156.2016.1235743>
- Mulenga, E. M., & Folokwe, K. (2025). Exploring Indigenous Teaching Approaches for Geometry Instruction in Three Secondary Schools Within Zambia. *Mathematics Teaching-Research Journal*, 17(1), 153–192.
- Munisamy, S., & Doraisamy, L. (1998). Levels of understanding of probability concepts among secondary school pupils. *International Journal of Mathematical Education in Science and Technology*, 29(1), 39–45. <https://doi.org/10.1080/0020739980290104>
- Murphy, S., MacDonald, A., Wang, C. A., & Danaia, L. (2019). Towards an Understanding of STEM Engagement: a Review of the Literature on Motivation and Academic Emotions. *Canadian Journal of Science, Mathematics and Technology Education*, 19(3), 304–320. <https://doi.org/10.1007/s42330-019-00054-w>
- Naimanova, D., Lebedeva, L., Akpayeva, A., Astambayeva, Z., Ryabova, E., & Yessenova, K. (2023). Investigation of Primary School Teachers’ Student-Centered Teaching and Technology Integration Competencies. *International Journal of Education in Mathematics, Science and Technology*, 11(6), 1386–1404. <https://doi.org/10.46328/ijemst.3681>
- Nakakoji, Y., Wilson, R., & Poladian, L. (2014). Mixed Methods Research on the Nexus Between Mathematics and Science. *International Journal of Innovation in Science and Mathematics Education*, 22(6).
- Nakypbek, Z. Y., Kainbaeva, L. S., Yensebayeva, G. M., Yergalauova, Z. A., Menlikhozhayeva, S. K., & Ibraev, S. S. (2025). Analysis of mathematical literacy and assessment literacy of mathematics preservice teachers. *Eurasia Journal of Mathematics, Science and Technology Education*, 21(12), em2748. <https://doi.org/10.29333/ejmste/17494>
- Nassir, L., Kozhasheva, G., Issayeva, G., Gavrilova, Y., & Yessenova, M. (2025). Cluster analysis of student satisfaction in a criteria-based assessment course with a project-based learning approach. *Eurasia Journal of Mathematics, Science and Technology Education*, 21(3), em2594. <https://doi.org/10.29333/ejmste/16032>
- Navarro, M., Gómez-Arízaga, M. P., Martin, A., Roa-Tampe, K., Conejeros-Solar, M. L., Kronborg, L., Valdivia-Lefort, M., Castillo-Hermosilla, H., & Rivera-Lino, B. (2020). Socio-emotional Dimensions in Gifted Chilean High School Students with Interests in STEM: Influence of Gender and University Enrichment Program Participation. *Eurasia Journal of Mathematics, Science and Technology Education*, 16(12), em1929. <https://doi.org/10.29333/ejmste/9374>
- Navarro, M., Martin, A., & Gómez-Arízaga, M. P. (2022). Profiles of Pre-Service Primary Teachers: Attitudes, Self-Efficacy, and Gender Stereotypes in Teaching Science and

- Mathematics. *Eurasia Journal of Mathematics, Science and Technology Education*, 18(1), em2062. <https://doi.org/10.29333/ejmste/11483>
- Navarro, M., Martin, A., Montoya, M. F., & Concha, S. (2024). Future primary teachers and pedagogical interactions with boys and girls. *Eurasia Journal of Mathematics, Science and Technology Education*, 20(5), em2443. <https://doi.org/10.29333/ejmste/14468>
- Navarro-Ibarra, L. A., Cuevas-Salazar, O., Acuña-Michel, L. L., & Valenzuela-Ochoa, J. M. (2025). Mathematics education and technology: Bibliometric analysis and systematic review (2000-2024). *Eurasia Journal of Mathematics, Science and Technology Education*, 21(3), em2607. <https://doi.org/10.29333/ejmste/16072>
- Ní Fhloinn, E., Fitzmaurice, O., Mac an Bhaird, C., & O'Sullivan, C. (2016). Gender Differences in the Level of Engagement with Mathematics Support in Higher Education in Ireland. *International Journal of Research in Undergraduate Mathematics Education*, 2(3), 297–317. <https://doi.org/10.1007/s40753-016-0031-4>
- NIELSEN, P. L., BEAN, N. W., & LARSEN, R. A. A. (2018). THE IMPACT OF A FLIPPED CLASSROOM MODEL OF LEARNING ON A LARGE UNDERGRADUATE STATISTICS CLASS. *STATISTICS EDUCATION RESEARCH JOURNAL*, 17(1), 121–140. <https://doi.org/10.52041/serj.v17i1.179>
- Nkechi, A., Azuka, B. F., Ajie, J. I., Oluwaniyi, S. D., Okwuoza, S. O., Durojaiye, D. S., Lemo, S. M., Ojo, S. G., Awogbemi, C. A., & Pelemo, B. J. (2018). Pilot study on the effect of the use of cultural materials and women's stories on the academic achievement of senior secondary school students in geometry in Abuja, Nigeria. *Mathematics Teaching-Research Journal*, 9(3-4), 3–21.
- Nolan, K. (2003). Casting shadows in the science classroom. *Canadian Journal of Science, Mathematics and Technology Education*, 3(2), 213–229. <https://doi.org/10.1080/14926150309556561>
- Norton, S. (2006). Pedagogies for the engagement of girls in the learning of proportional reasoning through technology practice. *Mathematics Education Research Journal*, 18(3), 69–99. <https://doi.org/10.1007/bf03217443>
- Norton, S. J., & Rennie, L. J. (1998). Students' attitudes towards mathematics in single-sex and coeducational schools. *Mathematics Education Research Journal*, 10(1), 16–36. <https://doi.org/10.1007/bf03217120>
- Nortvedt, G. A., & Buchholtz, N. (2018). Assessment in mathematics education: responding to issues regarding methodology, policy, and equity. *ZDM*, 50(4), 555–570. <https://doi.org/10.1007/s11858-018-0963-z>
- Nortvedt, G. A., & Siqveland, A. (2018). Are beginning calculus and engineering students adequately prepared for higher education? An assessment of students' basic mathematical knowledge. *International Journal of Mathematical Education in Science and Technology*, 50(3), 325–343. <https://doi.org/10.1080/0020739x.2018.1501826>

- Noyes, A. (2009). Exploring social patterns of participation in university-entrance level mathematics in England. *Research in Mathematics Education*, 11(2), 167–183. <https://doi.org/10.1080/14794800903063356>
- Noyes, A., & Adkins, M. (2016). Studying advanced mathematics in England: findings from a survey of student choices and attitudes. *Research in Mathematics Education*, 18(3), 231–248. <https://doi.org/10.1080/14794802.2016.1188139>
- Odutayo, A. O., & Fonseca, K. (2024). Making quadratic functions interesting: Students teams-achievement division instructional strategy. *Eurasia Journal of Mathematics, Science and Technology Education*, 20(1), em2386. <https://doi.org/10.29333/ejmste/14092>
- Oficiari, J. R., Reonisto, J., & Bactol, A. (2025). Well-Statement Problem: What is the Level of Problem-Posing Performance of Pre-Service Mathematics Teachers? *Mathematics Teaching-Research Journal*, 17(6), 223–247.
- Ojo, S. G. (2022). Effects of Animated Instructional Packages on Achievement and Interest of Junior Secondary School Student in Algebra. *Mathematics Teaching-Research Journal*, 14(1), 99–113.
- Oladejo, A. I., Olateju, T. T., Okebukola, P. A., Agboluaje, T. M., Sanni, R., Shabani, J., Akinola, V. O., & Ebisin, A. (2023). The convergence of culture, technology and context: A pathway to reducing Mathophobia and improving achievement in mathematics. *School Science and Mathematics*, 123(2), 82–96. <https://doi.org/10.1111/ssm.12573>
- Oladunni, M. O. (1998). An experimental study on the effectiveness of metacognitive and heuristic problem solving techniques on computational performance of students in mathematics. *International Journal of Mathematical Education in Science and Technology*, 29(6), 867–874. <https://doi.org/10.1080/0020739980290608>
- Olawale, B. E., & Hendricks, W. (2024). Mathematics teachers' self-efficacy beliefs and its relationship with teaching practices. *Eurasia Journal of Mathematics, Science and Technology Education*, 20(1), em2392. <https://doi.org/10.29333/ejmste/14123>
- Opie, O. N., Ovat, S. V., & Meremikwu, A. N. (2014). School Variables on Junior Secondary School Students' Mathematics Performance in Cross River State, Nigeria. *The International Journal of Science, Mathematics and Technology Learning*, 22(1), 1–9. <https://doi.org/10.18848/2327-7971/cgp/v22i01/49020>
- Opolot-Okurut, C. (2005). Student attitudes towards Mathematics in Ugandan secondary schools. *African Journal of Research in Mathematics, Science and Technology Education*, 9(2), 167–174. <https://doi.org/10.1080/10288457.2005.10740587>
- Opstad, L. (2021). Factors Explaining Business Student Attitudes Towards Mathematics: Does Gender Still Matter? *European Journal of Science and Mathematics Education*, 9(2), 13–25. <https://doi.org/10.30935/scimath/10771>
- Opyene-Eluk, P., & Opolot-Okurut, C. (1995). Gender and school-type differences in mathematics achievement of senior three pupils in central Uganda: an exploratory study.

International Journal of Mathematical Education in Science and Technology, 26(6), 871–886. <https://doi.org/10.1080/0020739950260608>

- Orhun, N. (2007). An investigation into the mathematics achievement and attitude towards mathematics with respect to learning style according to gender. *International Journal of Mathematical Education in Science and Technology*, 38(3), 321–333. <https://doi.org/10.1080/00207390601116060>
- Özgün-Koca, S. A., & Altay, M. K. (2009). An Investigation of Proportional Reasoning Skills of Middle School Students. *Investigations in Mathematics Learning*, 2(1), 26–48. <https://doi.org/10.1080/24727466.2009.11790289>
- Ozmantar, M. F., & Agac, G. (2023). Mathematics teacher educators' knowledge sources in teacher education practices. *Mathematics Education Research Journal*, 35(1), 175–201. <https://doi.org/10.1007/s13394-021-00382-x>
- Pampaka, M., & Williams, J. (2016). Mathematics teachers' and students' perceptions of transmissionist teaching and its association with students' dispositions. *Teaching Mathematics and its Applications*, 35(3), 118–130. <https://doi.org/10.1093/teamat/hrw007>
- Pampaka, M., Kleanthous, I., Hutcheson, G. D., & Wake, G. (2011). Measuring mathematics self-efficacy as a learning outcome. *Research in Mathematics Education*, 13(2), 169–190. <https://doi.org/10.1080/14794802.2011.585828>
- Pantaleon, K. V., Juniati, D., & Lukito, A. (2023). Female Student Mathematical Communication Ability in the Proving Process: a Review Based on Math Anxiety. *Bolema: Boletim de Educação Matemática*, 37(77), 1299–1316. <https://doi.org/10.1590/1980-4415v37n77a18>
- Papanastasiou, E. C., & Bottiger, L. (2004). *Math clubs and their potentials: making mathematics fun and exciting. A case study of a math club*. *International Journal of Mathematical Education in Science and Technology*, 35*(2), 159–171. <https://doi.org/10.1080/00207390310001638395>
- Parise, M. M. (2021). Gender, sex, and heteronormativity in high school statistics textbooks. *Mathematics Education Research Journal*, 33(4), 757–785. <https://doi.org/10.1007/s13394-021-00390-x>
- Park, H., Byun, S. Y., Sim, J., Han, H. S., & Baek, Y. S. (2016). Teachers' Perceptions and Practices of STEAM Education in South Korea. *EURASIA Journal of Mathematics, Science and Technology Education*, 12(7). <https://doi.org/10.12973/eurasia.2016.1531a>
- Park, J. H., & Han, S. (2023). Investigation of a developmental path model for interest in the study of mathematics. *Eurasia Journal of Mathematics, Science and Technology Education*, 19(7), em2288. <https://doi.org/10.29333/ejmste/13274>
- Pasha-Zaidi, N., & Afari, E. (2015). Gender in STEM Education: an Exploratory Study of Student Perceptions of Math and Science Instructors in the United Arab Emirates. *International Journal of Science and Mathematics Education*, 14(7), 1215–1231. <https://doi.org/10.1007/s10763-015-9656-z>

- Patahuddin, S. M., & Lowrie, T. (2018). Examining Teachers' Knowledge of Line Graph Task: a Case of Travel Task. *International Journal of Science and Mathematics Education*, 17(4), 781–800. <https://doi.org/10.1007/s10763-018-9893-z>
- Pataky, N., Hőgye-Nagy, Á., & Olajos, T. (2025). Exploring special non-cognitive factors related to young students' academic performance in STEM subjects. *Eurasia Journal of Mathematics, Science and Technology Education*, 21(5), em2626. <https://doi.org/10.29333/ejmste/16254>
- Patkin, D., & Dayan, E. (2013). The intelligence of observation: improving high school students' spatial ability by means of intervention unit. *International Journal of Mathematical Education in Science and Technology*, 44(2), 179–195. <https://doi.org/10.1080/0020739x.2012.703335>
- Patmaniar, P., Amin, S. M., & Sulaiman, R. (2021). STUDENTS' GROWING UNDERSTANDING IN SOLVING MATHEMATICS PROBLEMS BASED ON GENDER: ELABORATING FOLDING BACK. *Journal on Mathematics Education*, 12(3), 507–530. <https://doi.org/10.22342/jme.12.3.14267.507-530>
- Paul, S., Grundmeier, D., & Moore-Russo, D. (2024). 3D Manipulatives in Integral Calculus: Student Achievement and Confidence in Solids-Of-Revolution Tasks. *Investigations in Mathematics Learning*, 16(2), 100–119. <https://doi.org/10.1080/19477503.2024.2312097>
- Pedersen, I. F., & Haavold, P. Ø. (2023). Students' mathematical beliefs and motivation in the context of inquiry-based mathematics teaching. *International Journal of Mathematical Education in Science and Technology*, 54(8), 1649–1663. <https://doi.org/10.1080/0020739x.2023.2189171>
- Perry, C. A. (2010). Motivation and Attitude of Preservice Elementary Teachers toward Mathematics. *School Science and Mathematics*, 111(1), 2–10. <https://doi.org/10.1111/j.1949-8594.2010.00054.x>
- Peters, M. L. (2012). EXAMINING THE RELATIONSHIPS AMONG CLASSROOM CLIMATE, SELF-EFFICACY, AND ACHIEVEMENT IN UNDERGRADUATE MATHEMATICS: A MULTI-LEVEL ANALYSIS. *International Journal of Science and Mathematics Education*, 11(2), 459–480. <https://doi.org/10.1007/s10763-012-9347-y>
- Petty, T., Wang, C., & Harbaugh, A. P. (2013). Relationships Between Student, Teacher, and School Characteristics and Mathematics Achievement. *School Science and Mathematics*, 113(7), 333–344. <https://doi.org/10.1111/ssm.12034>
- Phage, I. B., Lemmer, M., & Hitge, M. (2017). Probing Factors Influencing Students' Graph Comprehension Regarding Four Operations in Kinematics Graphs. *African Journal of Research in Mathematics, Science and Technology Education*, 21(2), 200–210. <https://doi.org/10.1080/18117295.2017.1333751>
- Pham, H. T. H., Pham, Q. D., & Bui, C. K. (2023). Factors affecting the effectiveness of math teachers' integrated teaching in Vietnam high schools. *Eurasia Journal of Mathematics, Science and Technology Education*, 19(12), em2376. <https://doi.org/10.29333/ejmste/13894>

- Piatek-Jimenez, K. (2008). Images of mathematicians: a new perspective on the shortage of women in mathematical careers. *ZDM*, 40(4), 633–646. <https://doi.org/10.1007/s11858-008-0126-8>
- Picciarelli, V., Gennaro, M. D., Loconsole, A., & Stella, R. (1995). Development in mastering of rational number concepts and rational number operations: the role of gender, cognitive style and intellectual skills. *International Journal of Mathematical Education in Science and Technology*, 26(3), 407–416. <https://doi.org/10.1080/0020739950260309>
- Plangg, S., & Fuchs, K. J. (2022). A Gender-Related Analysis of a Robots' Math Class. *International Journal for Technology in Mathematics Education*, 29(3), 143–164. https://doi.org/10.1564/tme_v29.3.03
- Polaki, M. V., & Nenty, H. J. (2001). Gender Differences in Mathematics Performance Attributions Among First Year Students at National University of Lesotho: Implications for Access to and Performance in Mathematics. *African Journal of Research in Mathematics Science and Technology Education*, 5(1), 41–52. <https://doi.org/10.1080/10288457.2001.10756128>
- Polly, D., Wang, C., Petty, T., & Binns, I. (2022). Exploring the empirical connection between student, teacher, school, and district-level variables on fifth grade students' mathematics achievement. *School Science and Mathematics*, 122(3), 169–178. <https://doi.org/10.1111/ssm.12520>
- Polydoros, G. (2024). Math Anxiety in The Virtual Classroom During Covid-19 Pandemic and its Relationship to Academic Achievement. *Mathematics Teaching-Research Journal*, 16(4), 213–242.
- Prendergast, M., & O'Donoghue, J. (2014). Influence of gender, single-sex and co-educational schooling on students' enjoyment and achievement in mathematics. *International Journal of Mathematical Education in Science and Technology*, 45(8), 1115–1130. <https://doi.org/10.1080/0020739x.2014.904530>
- Prendergast, M., Breen, C., Bray, A., Faulkner, F., Carroll, B., Quinn, D., & Carr, M. (2018). Investigating secondary students beliefs about mathematical problem-solving. *International Journal of Mathematical Education in Science and Technology*, 49(8), 1203–1218. <https://doi.org/10.1080/0020739x.2018.1440325>
- Pringle, R. M., Brkich, K. M., Adams, T. L., West-Olatunii, C., & Archer-Banks, D. A. (2012). Factors Influencing Elementary Teachers' Positioning of African American Girls as Science and Mathematics Learners. *School Science and Mathematics*, 112(4), 217–229. <https://doi.org/10.1111/j.1949-8594.2012.00137.x>
- Przybyla-Kuchek, J. (2021). The possibilities of feminist poststructural discourse analysis as an approach to gender research in the mathematics classroom. *Mathematics Education Research Journal*, 33(4), 689–711. <https://doi.org/10.1007/s13394-020-00364-5>
- Quaye, J. (2014). Exploring students' attitudes towards mathematics and mathematical achievement in secondary schools in England: the role of social class, gender and ethnicity.

Research in Mathematics Education, 17(1), 59–60.
<https://doi.org/10.1080/14794802.2014.971340>

- Radević, L., & Milovanović, I. (2023). Current Trends in Math Anxiety Research: a Bibliometric Approach. *International Journal of Science and Mathematics Education*, 22(6), 1345–1362. <https://doi.org/10.1007/s10763-023-10424-4>
- Rahe, M., & Quaiser-Pohl, C. (2023). Can (perceived) mental-rotation performance mediate gender differences in math anxiety in adolescents and young adults? *Mathematics Education Research Journal*, 35(1), 255–279. <https://doi.org/10.1007/s13394-021-00387-6>
- Rahman, N. A., Rosli, R., Rambely, A. S., Siregar, N. C., Capraro, M. M., & Capraro, R. M. (2022). Secondary school teachers' perceptions of STEM pedagogical content knowledge. *Journal on Mathematics Education*, 13(1), 119–134. <https://doi.org/10.22342/jme.v13i1.pp119-134>
- Rainsford, E., & Foster, C. (2025). Confidence Assessment, Calibration, and Gender. *International Journal of Science and Mathematics Education*, 23(8), 4073–4098. <https://doi.org/10.1007/s10763-025-10615-1>
- Ramos, I., & Lambating, J. (1996). Gender Differences in Risk-Taking Behavior and their Relationship to SAT-Mathematics Performance. *School Science and Mathematics*, 96(4), 202–207. <https://doi.org/10.1111/j.1949-8594.1996.tb10225.x>
- Ramos, I., & Lambating, J. (1996). Risk Taking: Gender Differences and Educational Opportunity. *School Science and Mathematics*, 96(2), 94–98. <https://doi.org/10.1111/j.1949-8594.1996.tb15816.x>
- Real, I. I., & Carvalho, C. (2025). The influence of gender stereotypes on self-efficacy and mathematical anxiety in Peruvian students aspiring to STEM careers. *Journal on Mathematics Education*, 16(2), 407–422. <https://doi.org/10.22342/jme.v16i2.pp407-422>
- Rebhorn, L. S., & Miles, D. D. (1999). High-Stakes Testing: Barrier to Gifted Girls in Mathematics and Science? *School Science and Mathematics*, 99(6), 313–319. <https://doi.org/10.1111/j.1949-8594.1999.tb17490.x>
- Rech, J. F., Juhler, S. M., & Johnson, H. L. (1995). The effectiveness of videotapes in the individualized instruction of intermediate algebra and college algebra. *International Journal of Mathematical Education in Science and Technology*, 26(3), 463–472. <https://doi.org/10.1080/0020739950260314>
- Redmond, P., & Gutke, H. (2019). STEMming the Flow: Supporting Females in STEM. *International Journal of Science and Mathematics Education*, 18(2), 221–237. <https://doi.org/10.1007/s10763-019-09963-6>
- Reinhold, F., Hofer, S., Berkowitz, M., Strohmaier, A., Scheuerer, S., Loch, F., Vogel-Heuser, B., & Reiss, K. (2020). The role of spatial, verbal, numerical, and general reasoning abilities in complex word problem solving for young female and male adults. *Mathematics Education Research Journal*, 32(2), 189–211. <https://doi.org/10.1007/s13394-020-00331-0>

- Reinholz, D. L., & Shah, N. (2018). Equity Analytics: A Methodological Approach for Quantifying Participation Patterns in Mathematics Classroom Discourse. *Journal for Research in Mathematics Education*, 49(2), 140–177. <https://doi.org/10.5951/jresmetheduc.49.2.0140>
- Reinholz, D., Johnson, E., Andrews-Larson, C., Stone-Johnstone, A., Smith, J., Mullins, B., Fortune, N., Keene, K., & Shah, N. (2022). When Active Learning Is Inequitable: Women's Participation Predicts Gender Inequities in Mathematical Performance. *Journal for Research in Mathematics Education*, 53(3), 204–226. <https://doi.org/10.5951/jresmetheduc-2020-0143>
- Rennie, L. J., & Parker, L. H. (1997). Students' and teachers' perceptions of single-sex and mixed-sex mathematics classes. *Mathematics Education Research Journal*, 9(3), 257–273. <https://doi.org/10.1007/bf03217318>
- Reuter, D., & Reinhold, F. (2025). The interaction of task complexity and gender on mental rotation skills in primary school students. *International Electronic Journal of Mathematics Education*, 20(3), em0834. <https://doi.org/10.29333/iejme/16234>
- Rojas-Kramer, C. A., Escalera-Chávez, M. E., Moreno-García, E., & García-Santillán, A. (2016). Factors That Promote Anxiety toward Math on High School Students. *EURASIA Journal of Mathematics, Science and Technology Education*, 13(1). <https://doi.org/10.12973/eurasia.2017.00611a>
- Rolón, A. (2007). Gender differences in mathematics departments at colleges and universities across the United States: Towards an inclusive environment. *Mathematics Teaching-Research Journal*, 2(1).
- Rong, L., & Mononen, R. (2022). Error analysis of students with mathematics learning difficulties in Tibet. *Asian Journal for Mathematics Education*, 1(1), 52–65. <https://doi.org/10.1177/27527263221089357>
- Rosnelli, R., & Ristiana, P. A. (2023). Independent Curriculum Learning Management to Improve Students' Literacy and Numerical Competence in Schools. *International Journal of Education in Mathematics, Science and Technology*, 11(4), 946–963. <https://doi.org/10.46328/ijemst.3513>
- Ross, J. A., Scott, G., & Bruce, C. D. (2012). The Gender Confidence Gap in Fractions Knowledge: Gender Differences in Student Belief–Achievement Relationships. *School Science and Mathematics*, 112(5), 278–288. <https://doi.org/10.1111/j.1949-8594.2012.00144.x>
- Rothrock, K. S., & Gay, A. S. (2025). What College Freshmen Believe About Themselves: An Investigation of Mathematical Mindset, Identity, Self-Efficacy, and Use of Self-Regulated Learning Strategies in Mathematics. *International Journal of Research in Undergraduate Mathematics Education*, 11(2), 187–216. <https://doi.org/10.1007/s40753-023-00229-9>
- Rowe, W. R., & Easterday, K. E. (1987). An investigation of the effectiveness of the proposal of Richard A. Earle for teaching word problems in mathematics in selected topics and graded

- level. *International Journal of Mathematical Education in Science and Technology*, 18(3), 365–373. <https://doi.org/10.1080/0020739870180305>
- Rubel, L. H., & Nicol, C. (2020). The power of place: spatializing critical mathematics education. *Mathematical Thinking and Learning*, 22(3), 173–194. <https://doi.org/10.1080/10986065.2020.1709938>
- Rubel, L. H., Ayalon, M., & Shahbari, J. A. (2024). Sequencing & selecting solutions in a gendered world. *Mathematical Thinking and Learning*, 26(2), 185–207. <https://doi.org/10.1080/10986065.2022.2062641>
- Rubel, L. H., Herbel-Eisenman, B., Peralta, L. M., Lim, V., Jiang, S., & Kahn, J. (2022). Intersectional feminism to reenvision mathematical literacies & precarity. *Research in Mathematics Education*, 24(2), 224–248. <https://doi.org/10.1080/14794802.2022.2089908>
- Ruthven, K. (2001). Issues of cultural diversity in school mathematics. *International Journal of Mathematical Education in Science and Technology*, 32(3), 355–364. <https://doi.org/10.1080/00207390110040175>
- Ryan, V., Fitzmaurice, O., & O'donoghue, J. (2021). INVESTIGATING STUDENT KNOWLEDGE IN MATHEMATICS AT THE END OF THEIR FIRST YEAR OF POST-PRIMARY EDUCATION IN IRELAND: A CASE STUDY. *Teaching of Mathematics*, 24(2), 55–75.
- Ryan, V., O'Donoghue, J., & Fitzmaurice, O. (2022). Student Interest and Engagement in Mathematics After the First Year of Secondary Education. *European Journal of Science and Mathematics Education*, 10(4), 436–454. <https://doi.org/10.30935/scimath/12180>
- Sa'dijah, C., Murtafiah, W., Anwar, L., Nurhakiki, R., & Cahyowati, E. T. D. (2021). TEACHING HIGHER ORDER THINKING SKILLS IN MATHEMATICS CLASSROOMS: GENDER DIFFERENCES. *Journal on Mathematics Education*, 12(1), 159–180. <https://doi.org/10.22342/jme.12.1.13087.159-180>
- Saadati, F., Giaconi, V., & Barrera-Pedemonte, F. (2023). Mathematical problem-solving-related affect across gender and grade-level among upper primary students. *International Journal of Mathematical Education in Science and Technology*, 55(10), 2436–2457. <https://doi.org/10.1080/0020739x.2022.2159890>
- Saadati, F., Giaconi, V., Chandia, E., Fuenzalida, N., & Rodríguez Donoso, M. (2021). Beliefs and Practices About Remote Teaching Processes During the Pandemic: A Study with Chilean Mathematics Teachers. *Eurasia Journal of Mathematics, Science and Technology Education*, 17(11), em2023. <https://doi.org/10.29333/ejmste/11201>
- Sabbah, S., & Heyd-Metzuyanin, E. (2020). Integration of Arab Female Students at a Technological University—Narratives of Identity in Figured Worlds. *International Journal of Science and Mathematics Education*, 19(5), 977–996. <https://doi.org/10.1007/s10763-020-10084-8>
- Sahin, A., & Waxman, H. C. (2020). Characteristics of Secondary Students who have Intentions to Choose a STEM Major in College: Findings from a Three-Year Study. *Eurasia Journal*

of Mathematics, Science and Technology Education, 16(12), em1922.
<https://doi.org/10.29333/ejmste/9332>

- Sahin, A., Ekmekci, A., & Waxman, H. C. (2017). Collective Effects of Individual, Behavioral, and Contextual Factors on High School Students' Future STEM Career Plans. *International Journal of Science and Mathematics Education*, 16(S1), 69–89. <https://doi.org/10.1007/s10763-017-9847-x>
- Sahin, A., Waxman, H. C., Demirci, E., & Rangel, V. S. (2019). An Investigation of Harmony Public School Students' College Enrollment and STEM Major Selection Rates and Perceptions of Factors in STEM Major Selection. *International Journal of Science and Mathematics Education*, 18(7), 1249–1269. <https://doi.org/10.1007/s10763-019-10017-0>
- Said, Z., Mansour, N., & Abu-Tineh, A. (2023). Integrating technology pedagogy and content knowledge in Qatar's preparatory and secondary schools: The perceptions and practices of STEM teachers. *Eurasia Journal of Mathematics, Science and Technology Education*, 19(6), em2271. <https://doi.org/10.29333/ejmste/13188>
- Sánchez Aguilar, M., & Castaneda, A. (2019). A Foucauldian Analysis of Representations of Mathematicians in Lower Secondary Mexican Mathematics Textbooks. *International Journal of Science and Mathematics Education*, 18(4), 753–770. <https://doi.org/10.1007/s10763-019-09986-z>
- Šaparauskas, J., & Kačerauskas, T. (2017). Educational Choice Regarding Technical Education: Research with Case Study. *EURASIA Journal of Mathematics, Science and Technology Education*, 13(7). <https://doi.org/10.12973/eurasia.2017.00741a>
- Sasson, I. (2018). Participation in Research Apprenticeship Program: Issues Related to Career Choice in STEM. *International Journal of Science and Mathematics Education*, 17(3), 467–482. <https://doi.org/10.1007/s10763-017-9873-8>
- Saw, G. K., Kunisaki, L. T., Lin, S., Culbertson, R., & Megyesi-Brem, K. A. (2025). Opportunities for Deeper Learning and Adolescents' Creative Thinking Competency in STEM: The Mediating Role of Situated Expectancy-Value Beliefs. *International Journal of Science and Mathematics Education*, 23(6), 2089–2112. <https://doi.org/10.1007/s10763-025-10567-6>
- Sawaftah, M., & Al Mubasher, Z. (2021). The Impact of Using iPads in Teaching the 3D Shapes Textbook Unit and Promoting Spatial Visualization and Achievement in Mathematics among Eighth-Grade Students. *The International Journal of Science, Mathematics and Technology Learning*, 29(1), 29–47. <https://doi.org/10.18848/2327-7971/cgp/v29i01/29-47>
- Schinck, A. G., Neale, H. W., Pugalee, D. K., & Cifarelli, V. V. (2008). Using Metaphors to Unpack Student Beliefs About Mathematics. *School Science and Mathematics*, 108(7), 326–333. <https://doi.org/10.1111/j.1949-8594.2008.tb17845.x>
- Schwartz, C. S., & Sinicrope, R. (2013). Prospective Elementary Teachers' Perceptions of Gender Differences in Children's Attitudes Toward Mathematics. *School Science and Mathematics*, 113(8), 410–416. <https://doi.org/10.1111/ssm.12045>

- Seegers, G., & Boekaerts, M. (1996). Gender-Related Differences in Self-Referenced Cognitions in Relation to Mathematics. *Journal for Research in Mathematics Education*, 27(2), 215. <https://doi.org/10.2307/749601>
- Seki, J. M., & Menon, R. (2007). Incorporating Mathematics into the Science Program of Students Labeled “At-Risk”. *School Science and Mathematics*, 107(2), 61–69. <https://doi.org/10.1111/j.1949-8594.2007.tb17769.x>
- Sellami, A., Santhosh, M. E., Michaleczek, I., Alazaizeh, M., & Madad, J. (2024). Unveiling Teachers’ Instructional Self-efficacy in Science, Mathematics, and Technology: Personal and Contextual Influences. *Canadian Journal of Science, Mathematics and Technology Education*, 24(3-4), 418–438. <https://doi.org/10.1007/s42330-025-00359-z>
- Shah, J., Bather, J. R., Chen, Y., Kaul, S., Dias, J. J., & Goodman, M. S. (2024). Increasing Interest in Data Literacy: The Quantitative Public Health Data Literacy Training Program. *Journal of Statistics and Data Science Education*, 33(2), 210–222. <https://doi.org/10.1080/26939169.2024.2351559>
- Shahat, M. A., Al-Busaidi, F., Al-Ghafri, M., & Al Bahri, K. (2025). Navigating linguistic transitions: Pre-service science and math teachers’ perspectives on English as a medium of instruction in professional preparation. *Eurasia Journal of Mathematics, Science and Technology Education*, 21(5), em2640. <https://doi.org/10.29333/ejmste/16397>
- Shahbari, J. A., & Rubel, L. H. (2025). Problem Solving Strategies as Gendered: Views from Mathematics Teachers. *International Journal of Science and Mathematics Education*, 23(7), 2477–2500. <https://doi.org/10.1007/s10763-025-10565-8>
- Shapira-Lishchinsky, O., & Zavelevsky, E. (2019). Multiple Appearances of Parental Interactions and Math Achievement on TIMSS International Assessment. *International Journal of Science and Mathematics Education*, 18(1), 145–161. <https://doi.org/10.1007/s10763-018-09949-w>
- Sherawat, J., & Punia, P. (2022). Impact of adoption of Information and Communication Technologies (ICTs) in Teaching Mathematics to Intellectually Disabled Children. *Mathematics Teaching-Research Journal*, 14(1), 41–66.
- Shirvani, H. (2013). Math Anxiety and Math Teaching Anxiety Among College Students with Different Majors. *The International Journal of Science, Mathematics and Technology Learning*, 19(1), 11–18. <https://doi.org/10.18848/2327-7971/cgp/v19i01/49027>
- Shirvani, H. (2016). The Relationship between Student Mathematics Scores and Learning Styles. *The International Journal of Science, Mathematics and Technology Learning*, 23(4), 15–22. <https://doi.org/10.18848/2327-7971/cgp/v23i04/15-22>
- Shittu, M., & Alex, J. K. (2025). The 5E instructional model of constructivism in mathematics education: Teachers’ beliefs and classroom practices. *Eurasia Journal of Mathematics, Science and Technology Education*, 21(9), em2704. <https://doi.org/10.29333/ejmste/16884>
- Shores, M. L., & Shannon, D. M. (2007). The Effects of Self-Regulation, Motivation, Anxiety, and Attributions on Mathematics Achievement for Fifth and Sixth Grade Students. *School*

Science and Mathematics, 107(6), 225–236. <https://doi.org/10.1111/j.1949-8594.2007.tb18284.x>

Shultz, M., Close, E., Nissen, J., & Van Dusen, B. (2023). Enacting Culturally Relevant Pedagogy when “Mathematics Has No Color”: Epistemological Contradictions. *International Journal of Research in Undergraduate Mathematics Education*, 10(2), 486–515. <https://doi.org/10.1007/s40753-023-00219-x>

Sibanda, D., & Naidoo, J. (2023). Gender Gaps in Mathematics Achievement in TIMSS 2015: Evidence from Grade 9 South African Learners. *African Journal of Research in Mathematics, Science and Technology Education*, 27(2), 222–234. <https://doi.org/10.1080/18117295.2023.2257968>

Sievert, H., Hickendorff, M., van den Ham, A. K., & Heinze, A. (2025). Children’s Arithmetic Strategy Use and Strategy Change from Grade 3 to Grade 4. *International Journal of Science and Mathematics Education*, 23(8), 3577–3598. <https://doi.org/10.1007/s10763-025-10578-3>

Sikora, J., & Pitt, D. G. W. (2018). Does advanced mathematics help students enter university more than basic mathematics? Gender and returns to year 12 mathematics in Australia. *Mathematics Education Research Journal*, 31(2), 197–218. <https://doi.org/10.1007/s13394-018-0249-3>

Smail, L. (2016). USING BAYESIAN NETWORKS TO UNDERSTAND RELATIONSHIPS BETWEEN MATH ANXIETY, GENDERS, PERSONALITY TYPES, AND STUDY HABITS AT A UNIVERSITY IN JORDAN. *Journal on Mathematics Education*, 8(1). <https://doi.org/10.22342/jme.8.1.3405.17-34>

Smith, K. B., & Shotsberger, P. G. (1997). Assessing the Use of Graphing Calculators in College Algebra: Reflecting on Dimensions of Teaching and Learning. *School Science and Mathematics*, 97(7), 368–376. <https://doi.org/10.1111/j.1949-8594.1997.tb17378.x>

Solin, P. (2017). Bringing More STEAM to Mathematics Education. *International Journal for Technology in Mathematics Education*, 24(4), 191–198. https://doi.org/10.1564/tme_v24.4.03

Son, J. W., Han, S., Kang, C., & Kwon, O. N. (2016). A Comparative Analysis of the Relationship among Quality Instruction, Teacher Self-efficacy, Student Background, and Mathematics Achievement in South Korea and the United States. *EURASIA Journal of Mathematics, Science and Technology Education*, 12(7). <https://doi.org/10.12973/eurasia.2016.1532a>

Sondergeld, T. A., Provinzano, K., & Johnson, C. C. (2020). Investigating the impact of an urban community school effort on middle school STEM-related student outcomes over time through propensity score matched methods. *School Science and Mathematics*, 120(2), 90–103. <https://doi.org/10.1111/ssm.12387>

Souza, R. D. O., Gonçalves, T. O., & Moura-Silva, M. G. (2024). Mathematical anxiety is associated with the school context: correlational study between rural, urban and suburban

schools. *Bolema: Boletim de Educação Matemática*, 38. <https://doi.org/10.1590/1980-4415v38a210057>

- Sparks, D. M., & Pole, K. (2019). “Do we teach subjects or students?” Analyzing science and mathematics teacher conversations about issues of equity in the classroom. *School Science and Mathematics*, 119(7), 405–416. <https://doi.org/10.1111/ssm.12361>
- Spielman, L. J. (2008). Equity in mathematics education: unions and intersections of feminist and social justice literature. *ZDM*, 40(4), 647–657. <https://doi.org/10.1007/s11858-008-0113-0>
- Staus, N. L., Falk, J. H., Penuel, W., Dierking, L., Wyld, J., & Bailey, D. (2020). Interested, Disinterested, or Neutral: Exploring STEM Interest Profiles and Pathways in A Low-Income Urban Community. *Eurasia Journal of Mathematics, Science and Technology Education*, 16(6). <https://doi.org/10.29333/ejmste/7927>
- Steele, D. F., Levin, A. K., Blecksmith, R., & Shahverdian, J. (2005). The calculus of differences: Effects of a psychosocial, cultural, and pedagogical intervention in an all women’s university calculus class. *Mathematics Education Research Journal*, 17(3), 22–44. <https://doi.org/10.1007/bf03217420>
- Steinke, J., Applegate, B., Penny, J. R., & Merlino, S. (2021). Effects of Diverse STEM Role Model Videos in Promoting Adolescents’ Identification. *International Journal of Science and Mathematics Education*, 20(2), 255–276. <https://doi.org/10.1007/s10763-021-10168-z>
- Steinthorsdottir, O. B., & Sriraman, B. (2008). Exploring gender factors related to PISA 2003 results in Iceland: a youth interview study. *ZDM*, 40(4), 591–600. <https://doi.org/10.1007/s11858-008-0121-0>
- Steinthorsdottir, O. B., & Sriraman, B. (2009). Icelandic 5th-grade girls’ developmental trajectories in proportional reasoning. *Mathematics Education Research Journal*, 21(1), 6–30. <https://doi.org/10.1007/bf03217536>
- Stinson, D. W. (2013). Negotiating the “White Male Math Myth”: African American Male Students and Success in School Mathematics. *Journal for Research in Mathematics Education*, 44(1), 69–99. <https://doi.org/10.5951/jresmetheduc.44.1.0069>
- Stoehr, K. J. (2017). Building the wall brick by brick: one prospective teacher’s experiences with mathematics anxiety. *Journal of Mathematics Teacher Education*, 20(2), 119–139. <https://doi.org/10.1007/s10857-015-9322-y>
- Su, C. S., Díaz-Levicoy, D., & Hsu, C. C. (2024). The impact of a training cycle on statistics and probability for future primary schoolteachers with a gender-focused approach within the framework of education for sustainable development. *Eurasia Journal of Mathematics, Science and Technology Education*, 20(2), em2401. <https://doi.org/10.29333/ejmste/14177>
- Sumpter, L. (2015). ‘Boys Press All the Buttons and Hope It Will Help’: Upper Secondary School Teachers’ Gendered Conceptions About Students’ Mathematical Reasoning. *International Journal of Science and Mathematics Education*, 14(8), 1535–1552. <https://doi.org/10.1007/s10763-015-9660-3>

- Sumpter, L. (2015). Investigating Upper Secondary School Teachers' Conceptions: Is Mathematical Reasoning Considered Gendered? *International Journal of Science and Mathematics Education*, 14(S2), 347–362. <https://doi.org/10.1007/s10763-015-9634-5>
- Sumpter, L., & Sollerman, S. (2023). Changes in affect: patterns in Grade 4 and Grade 8 students' expressed emotional directions. *International Journal of Mathematical Education in Science and Technology*, 54(8), 1598–1613. <https://doi.org/10.1080/0020739x.2023.2184279>
- Süren, N., & Kandemir, M. A. (2020). The Effects of Mathematics Anxiety and Motivation on Students' Mathematics Achievement. *International Journal of Education in Mathematics, Science and Technology*, 8(3), 190. <https://doi.org/10.46328/ijemst.v8i3.926>
- Susanta, A., Susanto, E., Rusnilawati, R., Sumardi, H., & Ali, S. R. B. (2025). Literacy skills through the use of digital STEAM-inquiry learning modules: A comparative study of urban and rural elementary schools in Indonesia. *Eurasia Journal of Mathematics, Science and Technology Education*, 21(4), em2615. <https://doi.org/10.29333/ejmste/16170>
- Sutiarso, S., Coesamin, C., & Nurhanurawati, N. (2017). The Effect of Various Media Scaffolding on Increasing Understanding of Students' Geometry Concepts. *Journal on Mathematics Education*, 9(1). <https://doi.org/10.22342/jme.9.1.4291.95-102>
- Sutter, C. C., Givvin, K. B., Tucker, M. C., Givvin, K. A., Leandro-Ramos, A., & Solomon, P. L. (2022). Student Concerns and Perceived Challenges in Introductory Statistics, How the Frequency Shifted during COVID-19, and How They Differ by Subgroups of Students. *Journal of Statistics and Data Science Education*, 31(2), 188–200. <https://doi.org/10.1080/26939169.2022.2132325>
- Taasooobshirazi, G., Wagner, M., Brown, A., & Copeland, C. (2022). An Evaluation of College Students' Perceptions of Statisticians. *Journal of Statistics and Data Science Education*, 30(2), 138–153. <https://doi.org/10.1080/26939169.2022.2058655>
- Tao, S., Law, N. W. Y., & Ko, P. (2025). Parental Involvement and Students' STEM Self-efficacy in Hong Kong: A Multigroup Analysis of Gender and Age Disparities. *International Journal of Science and Mathematics Education*, 23(7), 3069–3091. <https://doi.org/10.1007/s10763-025-10568-5>
- Tariq, V. N., & Durrani, N. (2011). Factors influencing undergraduates' self-evaluation of numerical competence. *International Journal of Mathematical Education in Science and Technology*, 43(3), 337–356. <https://doi.org/10.1080/0020739x.2011.618552>
- Tariq, V. N., Qualter, P., Roberts, S., Appleby, Y., & Barnes, L. (2013). Mathematical literacy in undergraduates: role of gender, emotional intelligence and emotional self-efficacy. *International Journal of Mathematical Education in Science and Technology*, 44(8), 1143–1159. <https://doi.org/10.1080/0020739x.2013.770087>
- Tarr, J. E., Grouws, D. A., Chávez, Ó., & Soria, V. M. (2013). The Effects of Content Organization and Curriculum Implementation on Students' Mathematics Learning in Second-Year High School Courses. *Journal for Research in Mathematics Education*, 44(4), 683–729. <https://doi.org/10.5951/jresmetheduc.44.4.0683>

- Taylor McNeill, R., & Jefferson, A. (2024). “Prove Yourself”: Exploring Epistemological Values in Mathematics Department Support and Oppression of Black Women Faculty. *Journal for Research in Mathematics Education*, 55(1), 31–50. <https://doi.org/10.5951/jresmetheduc-2022-0102>
- Tefera, A., Atnafu, M., & Michael, K. (2021). The Relevance of Current Ethiopian Primary School Teacher Education Program for Pre-service Mathematics Teacher’s Knowledge and Teacher Educator’s Awareness about Mathematics Knowledge for Teaching. *Eurasia Journal of Mathematics, Science and Technology Education*, 17(5), em1964. <https://doi.org/10.29333/ejmste/10858>
- Tella, A. (2007). The Impact of Motivation on Student’s Academic Achievement and Learning Outcomes in Mathematics among Secondary School Students in Nigeria. *EURASIA Journal of Mathematics, Science and Technology Education*, 3(2). <https://doi.org/10.12973/ejmste/75390>
- Tella, A. (2011). An assessment of mathematics teachers’ Internet self-efficacy: implications on teachers’ delivery of mathematics instruction. *International Journal of Mathematical Education in Science and Technology*, 42(2), 155–174. <https://doi.org/10.1080/0020739x.2010.519798>
- Tempelaar, D. T., Gijsselaers, W. H., & van der Loeff, S. S. (2006). Puzzles in Statistical Reasoning. *Journal of Statistics Education*, 14(1). <https://doi.org/10.1080/10691898.2006.11910576>
- Thurm, D., Vandervieren, E., Moons, F., Drijvers, P., Barzel, B., Klinger, M., van der Ree, H., & Doorman, M. (2022). Distance mathematics education in Flanders, Germany, and the Netherlands during the COVID 19 lockdown—the student perspective. *ZDM – Mathematics Education*, 55(1), 79–93. <https://doi.org/10.1007/s11858-022-01409-8>
- Toney, A. F. (2014). Alternate Trajectories: Women Moving into Mathematics Education. *Investigations in Mathematics Learning*, 6(3), 29–50. <https://doi.org/10.1080/24727466.2014.11790334>
- Tossavainen, T., & Faarinen, E. C. (2019). Swedish Fifth and Sixth Graders’ Motivational Values and the Use of ICT in Mathematics Education. *EURASIA Journal of Mathematics, Science and Technology Education*, 15(12). <https://doi.org/10.29333/ejmste/108533>
- Turabay, G., Mailybaeva, G., Seitenova, S., Meterbayeva, K., Duisenbayev, A., & Ismailova, G. (2023). Analysis of Intercultural Communication Competencies in Prospective Primary School Teachers’ Use of Internet Technologies. *International Journal of Education in Mathematics, Science and Technology*, 11(6), 1537–1554. <https://doi.org/10.46328/ijemst.3795>
- Twohill, A., NicMhuirí, S., Harbison, L., & Karakolidis, A. (2022). Primary Preservice Teachers’ Mathematics Teaching Efficacy Beliefs: the Role Played by Mathematics Attainment, Educational Level, Preparedness to Teach, and Gender. *International Journal of Science and Mathematics Education*, 21(2), 601–622. <https://doi.org/10.1007/s10763-022-10259-5>

- Ubuz, B. (2011). Factors associated with success in a calculus course: an examination of personal variables. *International Journal of Mathematical Education in Science and Technology*, 42(1), 1–12. <https://doi.org/10.1080/0020739x.2010.500694>
- Ubuz, B., & Aydinyer, Y. (2017). Measuring striving for understanding and learning value of geometry: a validity study. *International Journal of Mathematical Education in Science and Technology*, 48(7), 1072–1086. <https://doi.org/10.1080/0020739x.2017.1327087>
- UITTO, A. (2014). INTEREST, ATTITUDES AND SELF-EFFICACY BELIEFS EXPLAINING UPPER-SECONDARY SCHOOL STUDENTS' ORIENTATION TOWARDS BIOLOGY-RELATED CAREERS. *International Journal of Science and Mathematics Education*, 12(6), 1425–1444. <https://doi.org/10.1007/s10763-014-9516-2>
- Uludüz, Ş. M., & Çalik, M. (2022). A Thematic Review of STEM Education for Disadvantaged Students. *Canadian Journal of Science, Mathematics and Technology Education*, 22(4), 938–958. <https://doi.org/10.1007/s42330-022-00247-w>
- UMUGIRANEZA, O., BANSILAL, S., & NORTH, D. (2022). ANALYSIS OF TEACHERS' CONFIDENCE IN TEACHING MATHEMATICS AND STATISTICS. *STATISTICS EDUCATION RESEARCH JOURNAL*, 21(3), 1. <https://doi.org/10.52041/serj.v21i3.422>
- Ursini, S., & Sánchez, G. (2008). Gender, technology and attitude towards mathematics: a comparative longitudinal study with Mexican students. *ZDM*, 40(4), 559–577. <https://doi.org/10.1007/s11858-008-0120-1>
- Uysal, S., & Clark, K. M. (2023). Identifying emergent themes of students' transition to university mathematics: a qualitative inquiry with a focus on affective factors. *Teaching Mathematics and its Applications: An International Journal of the IMA*, 43(3), 159–178. <https://doi.org/10.1093/teamat/hrad006>
- Vale, C. (2002). Girls back off mathematics again: the views and experiences of girls in computer-based mathematics. *Mathematics Education Research Journal*, 14(3), 202–218. <https://doi.org/10.1007/bf03217363>
- van den Heuvel-Panhuizen, M., Elia, I., & Robitzsch, A. (2015). Kindergartners' performance in two types of imaginary perspective-taking. *ZDM*, 47(3), 345–362. <https://doi.org/10.1007/s11858-015-0677-4>
- van Es, C., & Weaver, M. M. (2018). Race, Sex, and their Influences on Introductory Statistics Education. *Journal of Statistics Education*, 26(1), 48–54. <https://doi.org/10.1080/10691898.2018.1434426>
- Vanayan, M., White, N., Yuen, P., & Teper, M. (1997). Beliefs and Attitudes Toward Mathematics Among Third- and Fifth-Grade Students: A Descriptive Study. *School Science and Mathematics*, 97(7), 345–351. <https://doi.org/10.1111/j.1949-8594.1997.tb17375.x>
- Varsavsky, C., & Norton, P. (1995). Building up the self-confidence of mathematics students in non-science courses. *International Journal of Mathematical Education in Science and Technology*, 26(4), 517–521. <https://doi.org/10.1080/0020739950260405>

- Vazquez, S. R., Ermeling, B. A., & Ramirez, G. (2020). Parental Beliefs on the Efficacy of Productive Struggle and Their Relation to Homework-Helping Behavior. *Journal for Research in Mathematics Education*, 51(2), 179–203. <https://doi.org/10.5951/jresmetheduc-2020-0019>
- Verschaffel, L. (2023). Strategy flexibility in mathematics. *ZDM – Mathematics Education*, 56(1), 115–126. <https://doi.org/10.1007/s11858-023-01491-6>
- Wafa, M. N., Hussaini, S. A. M., & Pazhman, J. (2020). Evaluation of Students' Mathematical Ability in Afghanistan's Schools Using Cognitive Diagnosis Models. *Eurasia Journal of Mathematics, Science and Technology Education*, 16(6). <https://doi.org/10.29333/ejmste/7834>
- Walshaw, M. (2001). A Foucauldian Gaze on Gender Research: What Do You Do When Confronted with the Tunnel at the End of the Light? *Journal for Research in Mathematics Education*, 32(5), 471. <https://doi.org/10.2307/749802>
- Wang, Y., Qin, K., Luo, C., Yang, T., & Xin, T. (2022). Profiles of Chinese mathematics teachers' teaching beliefs and their effects on students' achievement. *ZDM – Mathematics Education*, 54(3), 709–720. <https://doi.org/10.1007/s11858-022-01353-7>
- Watson, C. N., Duran, P., Castillo, A., Fuller, E., Potvin, G., & Kramer, L. (2025). The supportive role of active learning in a calculus course on low precalculus proficiency students. *International Journal of Mathematical Education in Science and Technology*, 56(4), 561–582. <https://doi.org/10.1080/0020739x.2023.2255189>
- Watson, J. M. (1989). An exploration of gender differences in tertiary mathematics. *Mathematics Education Research Journal*, 1(1), 1–10. <https://doi.org/10.1007/bf03217198>
- Watson, J., Chick, H., & Callingham, R. (2014). Average: the juxtaposition of procedure and context. *Mathematics Education Research Journal*, 26(3), 477–502. <https://doi.org/10.1007/s13394-013-0113-4>
- Wedega, T. (2007). Gender perspectives in mathematics education: intentions of research in Denmark and Norway. *ZDM*, 39(3), 251–260. <https://doi.org/10.1007/s11858-007-0026-3>
- Weldeana, H. N. (2014). GENDER POSITIONS AND HIGH SCHOOL STUDENTS' ATTAINMENT IN LOCAL GEOMETRY. *International Journal of Science and Mathematics Education*, 13(6), 1331–1354. <https://doi.org/10.1007/s10763-014-9548-7>
- Wiebe, E., Unfried, A., & Faber, M. (2018). The Relationship of STEM Attitudes and Career Interest. *EURASIA Journal of Mathematics, Science and Technology Education*, 14(10). <https://doi.org/10.29333/ejmste/92286>
- Wieselmann, J. R., Roehrig, G. H., & Kim, J. N. (2020). Who succeeds in STEM? Elementary girls' attitudes and beliefs about self and STEM. *School Science and Mathematics*, 120(5), 297–308. <https://doi.org/10.1111/ssm.12407>
- Wiest, L. R., & Crawford-Ferre, H. G. (2023). Middle School girls' self-efficacy and motivation toward Mathematics before and after attending an all-girls summer Mathematics camp. *Review of Science, Mathematics and ICT Education*. <https://doi.org/10.26220/rev.4165>

- Wild, S., & Neef, C. (2024). Learning Strategies in Mathematics for Related Study Programs Focusing on Cooperative Education in Germany – Viewed from the Perspectives of the Academic Disciplines, Economics and Engineering, as well as Gender. *International Journal of Science and Mathematics Education*, 23(5), 1237–1258. <https://doi.org/10.1007/s10763-024-10513-y>
- Wilder, E. I., & West, R. K. (2024). The Mathematical Autobiographies of College Faculty Participating in a Quantitative Reasoning Faculty Development Program: Stories of Trauma and Triumph. *Adults Learning Mathematics International Journal*, 18(1), 29–47.
- Williams, K., White, S. L. J., & English, L. D. (2024). Profiles of general, test, and mathematics anxiety in 9- and 12-year-olds: relations to gender and mathematics achievement. *Mathematics Education Research Journal*, 37(1), 161–186. <https://doi.org/10.1007/s13394-024-00485-1>
- Wilson, T. M., & MacGillivray, H. L. (2007). Counting on the basics: mathematical skills among tertiary entrants. *International Journal of Mathematical Education in Science and Technology*, 38(1), 19–41. <https://doi.org/10.1080/00207390600819029>
- Winkelmann, H., Heuvel-Panhuizen, M., & Robitzsch, A. (2008). Gender differences in the mathematics achievements of German primary school students: results from a German large-scale study. *ZDM*, 40(4), 601–616. <https://doi.org/10.1007/s11858-008-0124-x>
- Wong, B., & Copsey-Blake, M. (2022). Pragmatic, Persistent, and Precarious: The Pathways of Three Minority Ethnic Women in STEM Higher Education. *International Journal of Science and Mathematics Education*, 21(7), 2123–2142. <https://doi.org/10.1007/s10763-022-10337-8>
- Wutsqa, D. U., Prihastuti, P. P., Fauzan, M., & Listyani, E. (2024). Radial Basis Function Neural Network with ensemble clustering for modeling mathematics achievement in Indonesia based on cognitive and non-cognitive factors. *Journal on Mathematics Education*, 15(3), 751–770. <https://doi.org/10.22342/jme.v15i3.pp751-770>
- Xenofontos, C. (2024). Gender representations in school mathematics: a study of primary textbooks in the Republic of Cyprus. *International Journal of Mathematical Education in Science and Technology*, 56(9), 1648–1668. <https://doi.org/10.1080/0020739x.2024.2337945>
- Xenofontos, C., & Hizli Alkan, S. (2022). “They’re coming into school hungry, they’re not ready to learn”. Scottish teachers’ perceptions of marginalization in school mathematics. *Eurasia Journal of Mathematics, Science and Technology Education*, 18(6), em2116. <https://doi.org/10.29333/ejmste/12071>
- Xu, L., Sun, J., van Driel, J., Hobbs, L., Millar, V., Kewalramani, S., Richards, G., & Aranda, G. (2025). A Conceptual Framework for Fostering Gender Equity in Early Years STEM Education. *International Journal of Science and Mathematics Education*, 23(7), 2951–2973. <https://doi.org/10.1007/s10763-025-10553-y>

- Xu, Z., & Qi, C. (2019). The Relationship between Teacher-student Relationship and Academic Achievement: The Mediating Role of Self-efficacy. *EURASIA Journal of Mathematics, Science and Technology Education*, 15(10). <https://doi.org/10.29333/ejmste/105610>
- Yang, D. C., Li, M. N., & Lin, C. I. (2007). A Study of the Performance of 5th Graders in Number Sense and its Relationship to Achievement in Mathematics. *International Journal of Science and Mathematics Education*, 6(4), 789–807. <https://doi.org/10.1007/s10763-007-9100-0>
- Yang, X. (2013). Investigation of Junior Secondary Students' Perceptions of Mathematics Classroom Learning Environments in China. *EURASIA Journal of Mathematics, Science and Technology Education*, 9(3). <https://doi.org/10.12973/eurasia.2013.935a>
- Yani T, A., Buyung, B., Rosmaiyadi, R., Marhayani, D. A., Nirawati, R., Triani, S. N., & Soeharto, S. (2025). Examining the relationship between mathematical literacy and digital literacy among pre-service mathematics teachers. *Eurasia Journal of Mathematics, Science and Technology Education*, 21(5), em2635. <https://doi.org/10.29333/ejmste/16392>
- Yasar, M. (2016). High School Students' Attitudes towards Mathematics. *EURASIA Journal of Mathematics, Science and Technology Education*, 12(4). <https://doi.org/10.12973/eurasia.2016.1571a>
- Yates, S. M. (2002). The influence of optimism and pessimism on student achievement in mathematics. *Mathematics Education Research Journal*, 14(1), 4–15. <https://doi.org/10.1007/bf03217113>
- Yavuz Mumcu, H., & Cansız Aktaş, M. (2019). Development of an attitude-towards-using-mathematics scale for high-school students and an analysis of student attitudes. *International Journal of Mathematical Education in Science and Technology*, 51(1), 3–25. <https://doi.org/10.1080/0020739x.2019.1679398>
- YOLCU, A. (2014). MIDDLE SCHOOL STUDENTS' STATISTICAL LITERACY: ROLE OF GRADE LEVEL AND GENDER. *STATISTICS EDUCATION RESEARCH JOURNAL*, 13(2), 118–131. <https://doi.org/10.52041/serj.v13i2.285>
- Yorulmaz, A., Altiner, E. Ç., & Çekirdekci, S. (2023). A Study of Number Sense and Metacognitive Awareness of Primary School Fourth Grade Students. *Mathematics Teaching-Research Journal*, 15(5), 5–29.
- Younes, R. G., Capraro, R. M., Capraro, M. M., Rosli, R., Lee, Y., Vela, K., & Bevan, D. (2021). Jack and Jill Went Up the Hill, but Jill Won Both Ways: The True Story about Differential Academic Achievement. *International Journal of Innovation in Science and Mathematics Education*, 28(4). <https://doi.org/10.30722/ijisme.28.04.004>
- Young, J., & Cunningham, J. A. (2020). Repositioning Black Girls in Mathematics Disposition Research: New Perspectives from QuantCrit. *Investigations in Mathematics Learning*, 13(1), 29–42. <https://doi.org/10.1080/19477503.2020.1827664>

- Yusuf, Y., Suyitno, H., Sukestiyarno, Y. L., & Isnarto (2019). The Influence of Statistical Anxiety on Statistic Reasoning of Pre-service Mathematics Teachers. *Bolema: Boletim de Educação Matemática*, 33(64), 694–706. <https://doi.org/10.1590/1980-4415v33n64a12>
- Zakariya, Y. F., & Wardat, Y. (2023). Job satisfaction of mathematics teachers: an empirical investigation to quantify the contributions of teacher self-efficacy and teacher motivation to teach. *Mathematics Education Research Journal*, 36(4), 791–813. <https://doi.org/10.1007/s13394-023-00475-9>
- Zhao, N., Teng, X., Li, W., Li, Y., Wang, S., Wen, H., & Yi, M. (2019). A path model for metacognition and its relation to problem-solving strategies and achievement for different tasks. *ZDM*, 51(4), 641–653. <https://doi.org/10.1007/s11858-019-01067-3>
- Zhou, G., He, L., Salinitri, G., & Xu, L. (2019). Female Underrepresentation in STEM Subjects: An Exploratory Study of Female High School Students in China. *EURASIA Journal of Mathematics, Science and Technology Education*, 16(1). <https://doi.org/10.29333/ejmste/109657>
- Zhumabay, N., Balta, N., Zhadrayeva, L., Yilmaz, S., & Nadelson, L. S. (2024). Exploring the triad of mathematical engagement: Perceived abilities, interests, and teacher-student dynamics. *Journal on Mathematics Education*, 15(4), 1053–1076. <https://doi.org/10.22342/jme.v15i4.pp1053-1076>
- Zohar, A., & Gershikov, A. (2007). Gender and Performance in Mathematical Tasks: Does the Context Make a Difference? *International Journal of Science and Mathematics Education*, 6(4), 677–693. <https://doi.org/10.1007/s10763-007-9086-7>